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VENTRICLE TETHERING

Abstract

A method comprises coupling a first anchor to an annulus of a heart valve that is between a heart atrium and a heart ventricle, and extending a portion of a tether within the heart ventricle from the first anchor to a portion of a ventricular heart wall of the heart ventricle between an apex and a basal portion of the ventricular heart wall. The method can include positioning a second anchor over an externally oriented surface of the portion of the ventricular heart wall, and coupling the tether to the second anchor while the second anchor is over the externally oriented surface of the ventricular heart wall.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Patent Application No. PCT/US2023/079598, filed Nov. 14, 2023, which claims the benefit of U.S. Patent Application No. 63/385,191, filed Nov. 28, 2022, the entire disclosures all of which are incorporated by reference for all purposes.

BACKGROUND

[0002] The present disclosure generally relates to the field of cardiac repairs, and more particularly to minimally invasive cardiac repair operations. Cardiac abnormalities can contribute to various heart conditions, adversely affecting supply of oxygenated blood to the rest of the body. Insufficient supply of oxygen can produce any number of symptoms, adversely affecting quality of life. Cardiac repair procedures can be performed to treat the abnormalities and alleviate the symptoms.

SUMMARY

[0003] Described herein are methods, devices and/or systems related to reducing or reversing heart ventricular remodeling. A tether system can be configured to couple a first portion of a heart to a second portion of the heart to facilitate reducing or reversing remodeling of a heart ventricle. In some instances, a tether system can be configured to couple a portion of a heart valve annulus to a portion of a ventricular heart wall. For example, a tether system can comprise a first anchor configured to be deployed to an atrioventricular valve annulus, a second anchor configured to be deployed to an externally oriented surface of the ventricular wall, and a tether comprising a portion configured to extend within the heart ventricle to couple to first and second anchors to one another. In some instances, a tether system can be configured to couple a portion of a septal wall to a portion of a ventricular heart wall. For example, a tether system can comprise a first anchor configured to be deployed to the septal wall, a second anchor configured to be deployed to an externally oriented surface of the ventricular heart wall, and a tether comprising a portion configured to extend within the heart ventricle to couple to first and second anchors to one another. In some instances, a first portion of a ventricular heart wall can be coupled to a second portion of a ventricular heart wall. For example, a tether system can comprise a first anchor configured to be deployed to an externally oriented surface of the first portion of the ventricular heart wall, a second anchor configured to be deployed to an externally oriented surface of the second portion of the ventricular heart wall, and a tether comprising a portion configured to extend within the heart ventricle to couple to first and second anchors to one another. In some instances, a tether system can be deployed to couple a papillary muscle to a ventricular heart wall, including an apical portion of the ventricular heart wall. For example, a tether system can comprise a first anchor configured to be deployed to the papillary muscle, a second anchor configured to be deployed to an externally oriented surface of the ventricular heart wall, and a tether comprising a portion configured to extend within the heart ventricle to couple the first and second anchors to one another.

[0004] Methods and structures disclosed herein for treating a patient also encompass analogous methods and structures performed on or placed on a simulated patient, which is useful, for example, for training; for demonstration; for procedure and/or device development; and the like. The simulated patient can be physical, virtual, or a combination of physical and virtual. A simulation can include a simulation of all or a portion of a patient, for example, an entire body, a portion of a body (e.g., thorax), a system (e.g., cardiovascular system), an organ (e.g., heart), or any combination thereof. Physical elements can be natural, including human or animal cadavers, or portions thereof; synthetic; or any combination of natural and synthetic. Virtual elements can be entirely in silica, or overlaid on one or more of the physical components. Virtual elements can be presented on any combination of screens, headsets, holographically, projected, loud speakers, headphones, pressure transducers, temperature transducers, or using any combination of suitable technologies.

[0005] For purposes of summarizing the disclosure, certain aspects, advantages and novel features have been described herein. It is to be understood that not necessarily all such advantages may be achieved in accordance with any particular example. Thus, the disclosed examples may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Various examples are depicted in the accompanying drawings for illustrative purposes and should in no way be interpreted as limiting the scope of the disclosure. In addition, various features of different disclosed examples can be combined to form additional examples, which are part of this disclosure. Throughout the drawings, reference numbers may be reused to indicate correspondence between reference elements. However, it should be understood that the use of similar reference numbers in connection with multiple drawings does not necessarily imply similarity between respective examples associated therewith. Furthermore, it should be understood that the features of the respective drawings are not necessarily drawn to scale, and the illustrated sizes thereof are presented for the purpose of illustration of inventive aspects thereof. Generally, certain of the illustrated features may be relatively smaller than as illustrated in some examples or configurations.

[0007] FIG. **1** provides a cutaway view of a heart, and a side view of a tether system deployed into the heart and configured to couple a portion of a heart valve annulus and a portion of a ventricular heart wall to one another, in accordance with one or more examples.

[0008] FIG. **2** provides a cutaway view of a heart, and a side view of another tether system deployed into the heart and configured to couple a portion of a heart valve annulus and a portion of a ventricular heart wall to one another, in accordance with one or more examples.

[0009] FIG. **3** provides a flow diagram of a process for deploying a tether system configured to couple a heart valve annulus and a ventricular heart wall to one another, in accordance with one or more examples.

[0010] FIG. **4** provides a cutaway view of a heart, and a side view of a tether system deployed into the heart and configured to couple a portion of a septal wall of a heart ventricle and a portion of a ventricular heart wall of the heart ventricle to one another, where the tether system comprises a tether configured to extend directly between the septal wall and the ventricular heart wall, in accordance with one or more examples.

[0011] FIG. **5** provides a cutaway view of a heart, and a side view of a tether system deployed into the heart and configured to couple a portion of a septal wall of a heart ventricle and a portion of a ventricular heart wall of the heart ventricle to one another, where the tether system comprises a tether configured to form a loop within the heart ventricle between the septal wall and the ventricular heart wall, in accordance with one or more examples.

[0012] FIG. **6** provides a cutaway view of a heart, and a side view of a tether system deployed into the heart and configured to couple a portion of a septal wall of a heart ventricle and a portion of a ventricular heart wall of the heart ventricle to one another, where the tether system comprises a tether configured to form a loop within the heart ventricle between the septal wall and the ventricular heart wall, in accordance with one or more examples.

[0013] FIG. **7** provides a flow diagram of a process for deploying a tether system configured to couple a portion of a septal wall of a heart ventricle and a portion of a ventricular heart wall of the heart ventricle to one another, in accordance with one or more examples.

[0014] FIG. **8** provides a cutaway view of a heart and a side view of a tether system deployed into the heart and configured to couple a papillary muscle of a heart ventricle and a ventricular heart wall of the heart ventricle to one another, in accordance with one or more examples.

[0015] FIG. **9** provides a flow diagram of a process for deploying a tether system configured to couple a papillary muscle of a heart ventricle and a ventricular heart wall of the heart ventricle to one another, in accordance with one or more examples.

[0016] FIG. **10** provides a cutaway view of a heart and a side view of a tether system deployed into the heart and configured to couple a first portion of a ventricular heart wall of a heart ventricle and a second portion of the ventricular heart wall to one another via a papillary muscle of the heart ventricle, in accordance with one or more examples.

[0017] FIG. **11** provides a flow diagram of a process for deploying a tether system configured to couple a first portion of a ventricular heart wall of a heart ventricle and a second portion of the ventricular heart wall to one another via a papillary muscle of the heart ventricle, in accordance with one or more examples.

[0018] FIG. **12** provides a perspective view of a tether delivery system, in accordance with one or more examples.

[0019] FIGS. **13**A, **13**B, **13**C, **13**D and **13**E provide cutaway views of the heart and various steps of deploying a tether using the tether delivery system described with reference to FIG. **12**, in accordance with one or more examples.

DETAILED DESCRIPTION

[0020] The headings provided herein are for convenience only and do not necessarily affect the scope or meaning of the claims.

[0021] Although certain preferred examples and examples are disclosed below, inventive subject matter extends beyond the specifically disclosed examples to other alternative examples and/or uses and to modifications and equivalents thereof. Thus, the scope of the claims that may arise herefrom is not limited by any of the particular examples described below. For example, in any method or process disclosed herein, the acts or operations of the method or process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various operations may be described as multiple discrete operations in turn, in a manner that may be helpful in understanding certain examples; however, the order of description should not be construed to imply that these operations are order dependent. Additionally, the structures, systems, and/or devices described herein may be embodied as integrated components or as separate components. For purposes of comparing various examples, certain aspects and advantages of these examples are described. Not necessarily all such aspects or advantages are achieved by any particular example. Thus, for example, various examples may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may also be taught or suggested herein.

[0022] Certain standard anatomical terms of location are used herein to refer to the anatomy of animals, and namely humans, with respect to the preferred examples. Although certain spatially relative terms, such as “outer,” “inner,” “upper,” “lower,” “below,” “above,” “vertical,” “horizontal,” “top,” “bottom,” and similar terms, are used herein to describe a spatial relationship of one device/element or anatomical structure to another device/element or anatomical structure, it is understood that these terms are used herein for ease of description to describe the positional relationship between element(s)/structures(s), as illustrated in the drawings. It should be understood that spatially relative terms are intended to encompass different orientations of the element(s)/structures(s), in use or operation, in addition to the orientations depicted in the drawings. For example, an element/structure described as “above” another element/structure may represent a position that is below or beside such other element/structure with respect to alternate

orientations of the subject patient or element/structure, and vice-versa.
[0023] FIG. **1** shows certain anatomical features of human vasculature, including various features of a human heart **1**. The heart **1** includes four chambers, namely the left atrium **2**, the left ventricle **3**, the right ventricle **4**, and the right atrium **5**. A wall of muscle, referred to as the septal wall **10**, separates the left atrium **2** and right atrium **5**, and the left ventricle **3** and right ventricle **4**. Blood flow through the heart **1** is at least partially controlled by four valves, the mitral valve **6**, aortic valve **7**, tricuspid valve **8**, and pulmonary valve. The mitral valve **6** separates the left atrium **2** and the left ventricle **3** and controls blood flow therebetween. The aortic valve **7** separates and controls blood flow between the left ventricle **3** and the aorta **12**. The tricuspid valve **8** separates the right atrium **5** and the right ventricle **4** and controls blood flow therebetween. The pulmonary valve separates the right ventricle **4** and the pulmonary trunk or artery, controlling blood flow therebetween.
[0024] In a healthy heart, the heart valves can properly open and close in response to a pressure gradient present during various stages of the cardiac cycle (e.g., relaxation and contraction) to at least partially control the flow of blood to a respective region of the heart and/or to blood vessels. Deoxygenated blood arriving from the rest of the body generally flows into the right side of the heart for transport to the lungs, and oxygenated blood from the lungs generally flows into the left side of the heart for transport to the rest of the body. During ventricular diastole, deoxygenated blood arrive in the right atrium **5** from the inferior vena cava and superior vena cava **16** to flow into the right ventricle **4**, and oxygenated blood arrive in the left atrium **2** from the pulmonary veins to flow into the left ventricle **3**. During ventricular systole, deoxygenated blood from the right ventricle **4** can flow into the pulmonary trunk for transport to the lungs (e.g., via the left and right pulmonary arteries), and oxygenated blood can flow from the left ventricle **3** to the aorta **12** for transport to the rest of the body.
[0025] The shape of a heart ventricle may change over time due to any number of conditions. Remodeling of the ventricular wall can progress over time due to volume overload, pressure overload, and/or myocardial injury. For example, mitral valve dysfunction can contribute to volume overload that is exerted on the left ventricular wall. Elevated stress on the left ventricular heart wall, such as during diastole, can cause the shape of the left ventricular wall to change. Remodeling of the ventricular wall can contribute to displacement of the papillary muscles and tenting of the atrioventricular valve. For example, remodeling of the left ventricular wall can cause displacement of one or more papillary muscles of the left ventricle and tenting of the mitral valve, such as during systole.
[0026] The present disclosure provides systems, devices, and methods relating to tether systems configured to be deployed into the heart to facilitate reversal of and/or reduction in the shape remodeling of a heart ventricle. Although the systems, devices, and methods described herein are referred primarily to as being used to deploy one or more tether systems into a left heart ventricle, it will be understood that the systems, devices, and methods can be applied to deploy one or more tether systems to a right ventricle.
[0027] In some instances, a tether system can be deployed into the heart to reverse and/or reduce remodeling a heart ventricular wall, such as a left ventricular wall. Reversal and/or reduction in remodeling of the heart ventricular wall can facilitate return of one or more papillary muscles to their respective original location. Reversal and/or reduction in remodeling of the left heart ventricular wall can facilitate return of one or more papillary muscles of the left ventricle to their original location. In some instances, relocating the papillary muscle can facilitate improvement in valve leaflet coaptation, for example reducing mitral valve regurgitation. In some instances, one or more tether systems described herein can be deployed to reduce or eliminate long term mitral valve regurgitation, such as in patients who have undergone multi-vessel coronary artery bypass graft (CABG) procedures and/or patients who experience moderate to severe ischemic functional mitral valve regurgitation.

[0028] In some instances, coupling a heart valve annulus to a portion of a ventricular heart wall can facilitate pulling the portion of the ventricular heart wall upward and/or inward. For example, a tether system can comprise a first anchor configured to be deployed to an atrioventricular valve annulus, a second anchor configured to be deployed to an externally oriented surface of the ventricular wall, and a tether comprising a portion configured to extend within the heart ventricle between the first and second anchors to couple the first and second anchors to one another.
[0029] In some instances, a portion of a septal wall can be coupled to a portion of a ventricular heart wall to facilitate pulling the portion of the ventricular heart wall inward and/or upward. For example, a tether system can comprise a first anchor configured to be deployed to the portion of the septal wall, a second anchor configured to be deployed to an externally oriented surface of the portion of the ventricular heart wall, and a tether comprising a portion configured to extend within the heart ventricle between the first and second anchors to couple the first and second anchors to one another.
[0030] In some instances, a first portion of a ventricular heart wall can be coupled to a second portion of the ventricular heart wall to facilitate pulling the second portion of the ventricular heart wall inward. The second portion of the ventricular heart wall can be at a lower position relative to that of the first portion, for example an inferior position relative to that of the first portion. A tether system can comprise a first anchor configured to be deployed to a first portion of a ventricular heart wall, a second anchor configured to be deployed to an externally oriented surface of a second portion of the ventricular heart wall, and a tether comprising a portion configured to extend within the heart ventricle between the first and second anchors to couple the first and second anchors to one another. The portion of the tether configured to be within the heart ventricle can form a partial loop around a papillary muscle of the heart ventricle to facilitate pulling the second portion of the ventricular heart wall inward. In some instances, the first anchor can be over an externally oriented surface of the first portion of the ventricular heart wall.
[0031] In some instances, one or more tether systems can be deployed to couple a papillary muscle to a ventricular heart wall. In some instances, a tether system can comprise a first anchor configured to be deployed to a papillary muscle, a second anchor configured to be deployed to an externally oriented surface of a portion of the ventricular heart wall, and a tether comprising a portion configured to extend within the heart ventricle between the first and second anchors to couple the first and second anchors to one another. In some instances, the first anchor can be coupled to an apical portion of the heart ventricular wall. The tether system can be configured to pull relocate the papillary muscle, such as pulling the papillary muscle and portion of the ventricular wall adjacent to the papillary muscle inward.
[0032] A procedure as described herein to improve or restore the shape of a ventricular heart wall can comprise deploying one or more tether systems as described herein. The one or more tether systems can be delivered while the heart is beating. In some instances, one or more tether systems can be delivered using a minimally invasive technique. For example, the procedure to deliver one or more tether systems can be minimally invasive beating heart procedure. Such procedures can be used to reduce and/or eliminate heart ventricular wall remodeling. Reducing and/or eliminating heart ventricular wall remodeling can improve mitral regurgitation, such as functional mitral regurgitation. The tether systems described herein can be delivered using procedures that can be performed off-pump, allowing for less invasive procedures and/or facilitating appropriate adjustments in lengths of the tether systems, such as compared to on-pump procedures.
[0033] As used herein, a basal portion of a ventricular heart wall can refer to a top third of the ventricular heart wall, such as a superior third of the ventricular heart wall. A mid-portion of a ventricular heart wall can refer to a middle third of the ventricular heart wall. An apical portion of a ventricular heart wall can refer to a lower third of the ventricular heart wall, such as an inferior third of the ventricular heart wall. A basal portion of a septal wall can refer to a top third of the septal wall, such as a superior third of the septal wall. A mid-portion of a septal wall can refer to a

middle third of the septal wall. An apical portion of a septal wall can refer to a lower third of the septal wall, such as an inferior third of the septal wall.

[0034] Although tether systems are described herein primarily as comprising a suture and/or cord and a bulky-knot type anchor, any number of different mechanical tethering components can be applied. In some instances, one or more portions of the tether system, such as a tether and/or anchor portion, can comprise a material that is rigid, semi-rigid or flexible. In some instances, a tether of the tether system can comprise a length at least about twice as long as the tether is wide. For example, a tether tail of the tether can comprise a length at least about twice as long as the tether tail is wide. In some instances, an anchor, including a first and/or second anchor as described herein, can comprise any number of mechanical anchors, such as a T-fastener and/or hook. In some instances, a tether system can comprise a solid rod, including a rod comprising at least a portion that is rigid, semirigid, or flexible, and respective mechanical anchor coupled to each of the two distal end portions of the solid rod. For example, a tether system can comprise a T-fastener coupled to each of two distal end portions of a solid rod. It will be understood that the anchors described herein can be deployed in any order.

[0035] The methods, operations, steps, etc. described herein can be performed on a living animal or on a non-living cadaver, cadaver heart, anthropomorphic ghost, simulator (e.g., with the body parts, tissue, etc. being simulated), etc. For example, methods for treating a patient include methods for simulating the treatment on a simulated patient or anthropogenic ghost, which can include any combination of physical and virtual elements. Examples of physical elements include human or animal cadavers; any portions thereof, including organ systems, whole organs, or tissue; and manufactured elements, which can simulate the appearance, texture, resistance, or other characteristic. Virtual elements can include visual elements provided on a screen, or projected on a surface or volume, including virtual reality and augmented reality implementations. Virtual elements can also simulate other sensory stimuli, including sound, feel, and/or odor.

[0036] Any of the various systems, devices, apparatuses, etc. in this disclosure can be sterilized (e.g., with heat, radiation, ethylene oxide, hydrogen peroxide, etc.) to ensure they are safe for use with patients, and the methods herein can comprise sterilization of the associated system, device, apparatus, etc. (e.g., with heat, radiation, ethylene oxide, hydrogen peroxide, etc.).

[0037] FIGS. **1** and **2** show tether systems **100**, **200** deployed in the heart **1** for coupling a portion of a heart valve annulus and a portion of a ventricular heart wall **17** to one another. Each of the tether systems **100**, **200** can comprise a first anchor **110**, **210**, a second anchor **130**, **230**, and a tether **150**, **250** configured to couple the respective first anchor **110**, **210** and the second anchor **130**, **230** to one another. The first anchor **110**, **210** can be configured to be coupled to an annulus of a heart valve. In some instances, the heart valve can be between a heart atrium and a heart ventricle. For example, the first anchor **110**, **210** can be coupled to a heart valve between the left atrium **2** and the left ventricle **3**, such as the mitral valve **6**. The second anchor **130**, **230** can be configured to be positioned over an externally oriented surface of a portion of a ventricular heart wall **17** of the heart ventricle, such as a ventricular heart wall **17** of the left ventricle **3**. The second anchor **130**, **230** can be positioned over the externally oriented surface between an apex and a basal portion of the ventricular heart wall **17**. In some instances, the tether systems **100**, **200** can be configured to couple an annulus of a heart valve controlling blood flow into a heart ventricle to a portion of a ventricular heart wall of the heart ventricle so as to pull the portion of the ventricular heart wall upward toward the annulus. Relocating the portion of the ventricular heart wall upward and/or inward toward the annulus can reduce or reverse remodeling of the portion of the ventricular heart wall. For example, the tether systems **100**, **200** can be configured to couple an annulus of the mitral valve **6** to a portion of the ventricular heart wall **17** of the left ventricle **3**.

[0038] The position on the annulus to which a first anchor is deployed can be selected to provide the desired ventricular wall remodeling. In some instances, for example as shown in FIG. **1**, the first anchor **110** can be deployed to a portion of the annulus adjacent to a posterior leaflet **64** of the mitral valve **6**. Coupling a portion of the ventricular heart wall to a portion of the annulus adjacent to a posterior leaflet **64** of the mitral valve **6** can facilitate pulling the portion of the ventricular heart wall upward toward the portion of the annulus. In some instances, for example as shown in FIG. **2**, the first anchor **210** can be deployed to a portion of the annulus adjacent to an anterior leaflet **62** of the mitral valve **6**. Coupling a portion of the ventricular heart wall to a portion of the annulus adjacent to an anterior leaflet **62** of the mitral valve **6** can facilitate pulling the portion of the ventricular heart wall upward and inward toward the portion of the annulus.

[0039] In some instances, the portion of the ventricular heart wall over which the second anchor **130**, **230** is placed can be in a mid-portion of the ventricular heart wall. In some instances, the portion of the ventricular heart wall over which the second anchor **130**, **230** is placed can be between an apical portion and a mid-portion of the ventricular heart wall. In some instances, the portion of the ventricular heart wall over which the second anchor **130**, **230** is placed can be adjacent to a portion of the ventricular heart wall from which a papillary muscle extends. In some instances, the portion of the ventricular heart wall over which the second anchor **130**, **230** is placed can be adjacent to a portion of the ventricular heart wall from which an anterolateral papillary muscle extends.

[0040] The first anchor **110**, **210** can be configured to be disposed over an atrial facing portion of the portion of the heart valve annulus, such as the annulus of the mitral valve **6**. As described herein, the second anchor **130**, **230** can be configured to be positioned over an externally oriented surface of a ventricular heart wall, such as the ventricular heart wall **17** of the left ventricle **3**. In some instances, each of the tethers **150**, **250** can comprise a first distal portion **152**, **252** configured to extend through the heart valve annulus, such as from a ventricular facing surface to the atrial facing surface of the annulus. The first distal portion **152**, **252**, such as a first distal end **156**, **256**, of the tether **150**, **250** can be coupled to the first anchor **110**, **210**. The tether **150**, **250** can comprise another portion configured to be disposed through the ventricle, such as the left ventricle **3**. For example, a medial portion **158**, **258** of the tether **150**, **250** can be configured to be disposed within the left ventricle **3**. A second distal portion **154**, **254** can be configured to extend through the ventricular heart wall **17**, such as from a ventricular facing surface to the externally oriented surface such that the tether **150**, **250** can be coupled to the second anchor **130**, **230**. For example, the second distal portion **154**, **254** can extend through an incision and/or opening **40** made in the ventricular heart wall **17**.

[0041] In some instances, the second anchor **130**, **230** can comprise a first surface **132**, **232** configured to be oriented toward the ventricular heart wall and a second surface **134**, **234** configured to be oriented away from the ventricular heart wall. In some instances, the second anchor **130**, **230** can comprise a pad configuration. In some instances, the second anchor **130**, **230** can comprise a support platform configured to facilitate maintaining the second distal portion **154**, **254** of the tether **150**, **250** at the desired position on the ventricular heart wall. The first surface **132**, **232** can be configured to be over and in contact with the ventricular heart wall. In some instances, the second anchor **130**, **230** can comprise a pad configuration. In some instances, the second anchor **130**, **230** can comprise a support platform configured to facilitate distribution of the tension exerted by the tether **150**, **250** upon the ventricular heart wall to improve the reduction in remodeling achieved. For example, the support platform can facilitate distribution of the tension exerted by the tether **150**, **250** upon the ventricular heart wall over a larger area to allow a larger area of the ventricular heart wall to be reverse remodeled. The second anchor **130**, **230** can be a support platform having a first surface **132**, **232** configured to be over and in contact with the ventricular heart wall **17** and a second opposing surface **134**, **234** configured to be oriented away from the from the ventricular heart wall **17**.

[0042] In some instances, the first anchor **110**, **210** and the tether **150**, **250** can be an integral piece. In some instances, the first anchor **110**, **210** and the tether **150**, **250** can be formed using the same material. The first anchor **110**, **210** and the tether **150**, **250** can each comprise a respective portion

of a surgical cord and/or suture. For example, a portion of a suture **102**, **202** can form the first anchor **110**, **210**. Another portion of the suture **102**, **202** can form the tether **150**, **250**. As described in further detail herein, in some instances, the first anchor **110**, **210** can comprise a bulky-knot. For example, a portion of the suture **102**, **202** can be deployed through the portion of the heart valve annulus to form the first anchor **110**, **210** over and in contact with an atrium-facing surface of the annulus. Another portion of the suture **102**, **202** can form the tether **150**, **250**, for example extending through the annulus, the ventricle, and a portion of the ventricular wall, such that the suture **102**, **202** can be coupled to the second anchor **130**, **230** positioned over the externally oriented surface of the ventricular wall. For example, the first anchor **110**, **210** can be formed over a surface of the mitral valve annulus oriented toward the left atrium. The tether **150**, **250** can extend through the mitral valve annulus, left ventricle and a portion of the left heart ventricular wall adjacent to the anterolateral papillary muscle to couple to the second anchor **130**, **230**. The second anchor **130**, **230** can be positioned over the portion of the ventricular heart wall adjacent to a portion of the heart ventricular wall from which an anterolateral papillary muscle extends.
[0043] In some instances, each of the tethers **150**, **250** can comprise respective tether tails **160**, **260**. For example, the tethers **150**, **250** can each comprise a pair of tether tails **160***a*, **160***b*, **260***a*, **260***b*. For example, the tether tails **160***a*, **160***b*, **260***a*, **26***ob* can comprise a respective first distal portion **162***a*, **162***b*, **262***a*, **262***b* configured to extend through the portion of the annulus. In some instances, first distal ends **166***a*, **166***b*, **266***a*, **266***b* can be coupled to the first anchors **110**, **210**. A second distal portion **164***a*, **164***b*, **264***a*, **264***b* of each of the tether tails **160***a*, **160***b*, **260***a*, **260***b* can be configured to extend through the portion of the ventricular heart wall. In some instances, the second distal portions **164***a*, **164***b*, **264***a*, **264***b* can be configured to be coupled to the respective second anchors **130**, **230**. For example, the second distal portions **164***a*, **164***b*, **264***a*, **264***b* can be configured to extend through the respective second anchors **130**, **230** and form the knot **170**, **270** for coupling the tethers **150**, **250** to the second anchors **130**, **230**. A medial portion **168***a*, **168***b*, **268***a*, **268***b* of the tether tails **160***a*, **160***b*, **260***a*, **26***ob* can be configured to be disposed within the heart ventricle.
[0044] In some instances, the tether systems **100**, **200** can optionally comprise an elongate tube **180**, **280** comprising a lumen **182**, **282** configured to receive at least a portion of the tether **150**, **250** extending within the heart ventricle. The elongate tube **180**, **280** can be configured to prevent or reduce damage to heart tissue due to contact between the tether **150**, **250** and the heart tissue.
[0045] It will be understood that a tether system as described herein can be applied to either atrioventricular heart valve. For example, a first anchor can be deployed to be coupled to a tricuspid valve. A second anchor can be deployed to an externally oriented surface portion of a right ventricular heart wall. A tether can comprise a portion configured to extend within the right ventricle to couple the first anchor and the second anchor to one another. The tether system for coupling the tricuspid valve annulus to the right ventricular heart wall can comprise one or more features of the tether systems **100**, **200** described with reference to FIGS. **1** and **2**.
[0046] FIG. **3** is a process flow diagram showing an example of a process **300** for deploying a tether system as described herein, including one or more of the tether systems **100**, **200** described with reference to FIGS. **1** and **2**. In block **302**, the process can involve coupling a first anchor to an annulus of a heart valve that is between a heart atrium and a heart ventricle. In block **304**, the process can involve extending a portion of a tether within the heart ventricle from the first anchor to a portion of a ventricular heart wall of the heart ventricle between an apex and a basal portion of the ventricular heart wall. In block **306**, the process can involve positioning a second anchor over an externally oriented surface of the portion of the ventricular heart wall. In block **308**, the process can involve coupling the tether to the second anchor while the second anchor is over the externally oriented surface of the ventricular heart wall.
[0047] In some instances, extending the portion of the tether from the first anchor to the portion of the heart ventricular heart can comprise extending the tether from the first anchor to a mid-portion

of the ventricular heart wall. In some instances, extending the portion of the tether from the first anchor to the portion of the heart ventricular heart can comprise extending the tether from the first anchor to a portion of the ventricular heart wall between an apical portion and a mid-portion of the ventricular heart wall. In some instances, extending the portion of the tether from the first anchor to the portion of the ventricular heart wall can comprise extending the tether from the first anchor to a portion of the ventricular heart wall adjacent to a papillary muscle.

[0048] In some instances, coupling the first anchor to the annulus of the heart valve can comprise coupling the first anchor to a mitral valve annulus. Extending the portion of the tether within the heart ventricle from the first anchor to the portion of the ventricular heart wall of the heart ventricle between the apex and the basal portion of the ventricular wall can comprise extending the portion of the tether within the left heart ventricle from the first anchor coupled to the mitral valve annulus. Positioning the second anchor over the externally oriented surface of the portion of the ventricular heart wall can comprise positioning the second anchor over an externally oriented surface of a portion of a ventricular heart wall of the left heart ventricle. In some instances, coupling the tether to the second anchor while the second anchor is over the externally oriented surface of the ventricular heart wall can comprise coupling the tether to the second anchor while the second anchor is over the externally oriented surface of the ventricular heart wall of the left heart ventricle.

[0049] In some instances, coupling the first anchor to the annulus of the heart valve can comprise coupling the first anchor to a portion of the mitral valve annulus adjacent to a posterior leaflet. Alternatively, coupling the first anchor to the annulus of the heart valve can comprise coupling the first anchor to a portion of the mitral valve annulus adjacent to an anterior leaflet. For example, extending the portion of the tether from the first anchor to the portion of the ventricular heart wall can comprise extending the tether from the first anchor to a portion of the ventricular heart wall adjacent to anterolateral papillary muscle.

[0050] As described herein, a tether system can comprise an elongate tube configured to receive at least a portion of the portion of the tether extending from the first anchor to the second anchor, including at least a portion of the tether configured to be positioned within the ventricle. In some instances, the tether system deployment process can involve advancing an elongate tube along at least a portion of the tether within the heart ventricle. The elongate tube can comprise a central lumen configured to receive at least a portion of the tether that is disposed within the heart ventricle. For example, the elongate tube can be advanced along at least a portion of the tether disposed within the heart ventricle, after the first anchor is deployed to its target site and the tether is extended from the first anchor. In some instances, the elongate tube can be configured to extend along an entire or substantially entire length of the tether disposed within the heart ventricle. For example, the tether system deployment process can involve advancing an elongate tube along the entire or substantially entire portion of the tether that is disposed within the heart ventricle.

[0051] FIGS. **4**, **5** and **6** show examples of tether systems **400**, **500**, **600** configured to couple a septal wall **10** of a heart **1** and a ventricular heart wall **17**. Each of the tether systems **400**, **500**, **600** can comprise a first anchor **410**, **510**, **610** configured to be coupled to a portion of the septal wall **10** and a second anchor **430**, **530**, **630** configured to be coupled to a portion of the ventricular heart wall **17**. A tether **450**, **550**, **650** can extend between the respective first anchors **410**, **510**, **610** and second anchors **430**, **530**, **630**. Each tether **450**, **550**, **650** can be configured to couple the respective first anchors **410**, **510**, **610** and second anchors **430**, **530**, **630** to one another. For example, each of the tethers **450**, **550**, **650** can comprise a portion configured to be disposed through the heart ventricle between the first anchors **410**, **510**, **510** and second anchors **430**, **530**, **630**. Coupling the ventricular heart wall **17** to the septal wall **10** can be configured to pull the ventricular heart wall **17** inwards and/or upwards. The portion of the ventricular heart wall **17** can be pulled toward the portion of the septal wall **10**. For example, locations on the septal wall **10** to which the first anchors **410**, **510**, **610** are coupled and/or locations on the ventricular heart wall **17** to which the second anchors **430**, **530**, **630** are coupled can be selected such that desired relocation of the target portion of the ventricular heart wall **17** inwards and/or upwards can be achieved. In some instances, the first anchors **410**, **510**, **610** can be configured to be coupled to a basal portion of the septal wall **10**. In some instances, the first anchors **410**, **510**, **610** can be configured to be coupled to a mid-portion of the septal wall **10**. In some instances, the second anchors **430**, **530**, **630** can be configured to be coupled to a mid-portion of the ventricular heart wall **17**.

[0052] Each of the tethers **450**, **550**, **650** can comprise a first distal portion **452**, **552**, **652** and a second distal portion **454**, **554**, **654**. The first distal portions **452**, **552**, **652** can be configured to be disposed through at least a portion of the septal wall **10** such that the tethers **450**, **550**, **650** can be coupled to the first anchors **410**, **510**, **610**. For example, a first distal end **456**, **556**, **656** of the tether **450**, **550**, **650** can be coupled to the first anchor **410**, **510**, **610**. As described in further detail herein, in some instances, a first anchor can be configured to be disposed over a surface of a septal wall. Alternatively, a first anchor can be at least partially, including completely, embedded within a septal wall. FIGS. **4** and **6** show the first anchors **410**, **610** disposed over and/or in contact with a surface of the septal wall **10** having an orientation opposite relative to that oriented toward the heart ventricle through which the tethers **450**, **650** extend. The first anchors **410**, **510** can be configured to be disposed within a heart ventricle different from that within which the portion of the tethers **450**, **650** are configured to be disposed. The first anchors **410**, **610** are shown in FIGS. **4** and **6** as being disposed over and/or in contact with a surface of the septal wall **10** that is oriented toward a right heart ventricle **4**. The first distal portions **452**, **652** can comprise at least a portion configured to be disposed through an entire thickness of the septal wall **10** between the right and left ventricles **4**, **3** such that the tethers **450**, **650** can be coupled to the first anchors **410**, **610**. The first distal portion **152**, **252**, such as a first distal end **156**, **256**, of the tether **150**, **250** can be coupled to the first anchor **110**, **210**. FIG. **5** shows the first anchor **510** embedded within the septal wall **10**. The first distal portion **552** of the tether **550** can be configured to be disposed through only a portion of a thickness of the septal wall **10**. A medial portion **458**, **558**, **658** of the tethers **450**, **550**, **650** can be disposed within the left ventricle **3**.

[0053] Each of the second anchors **430**, **530**, **630** can be configured to be disposed over an externally oriented surface of the portion of the ventricular heart wall **17**. The second distal portions **454**, **554**, **654** can comprise at least a portion configured to extend through the portion of the ventricular heart wall **17** to couple to the second anchors **430**, **530**, **630**. FIGS. **4**, **5** and **6** show the second anchors **430**, **530**, **630** disposed over an externally oriented surface of a portion of the ventricular heart wall **17** of the left ventricle **3**. The second distal portions **454**, **554**, **654** can comprise at least a portion configured to extend through the portion of the ventricular heart wall **17** of the left ventricle **3** to couple to the second anchors **430**, **530**, **630**. The tethers **450**, **550**, **650** can comprise a respective portion configured to extend through each of at least a portion of the septal wall **10**, through the left heart ventricle **3** and through the ventricular heart wall **17** of the left heart ventricle **3**.

[0054] FIG. **4** shows the tether **450** being configured to extend directly between the first anchor **410** and the second anchor **430**. For example, the tether **450** can extend between the first anchor **410** and the second anchor **430** without forming any loops. The tether **450** can extend between the portion of the septal wall **10** and the portion of the ventricular heart wall **17** of the left ventricle **3** without forming any loops. FIGS. **5** and **6** show the tethers **550**, **650** comprising a portion configured to form one or more loops within the heart ventricle around papillary muscles of the heart ventricle and/or chordae tendineae coupled to the papillary muscles. The tethers **550**, **650** can comprise a portion configured to form a loop around respective portions of the papillary muscles and chordae tendineae in the heart ventricle. Medial portions **558**, **658** of the tethers **550**, **650** disposed within the left ventricle **3** can be configured to form a loop around one or more of the papillary muscles **19**, **21** in the left ventricle **3** and/or chordae tendineae **20**, **22** coupled to the papillary muscles **19**, **21**. For example, medial portions **558**, **658** of the tethers **550**, **650** can be configured to form a loop around both a posteromedial papillary muscle **21** or chordae tendineae **22** coupled to the posteromedial papillary muscle **21**, and an anterolateral papillary muscle **19** or chordae tendineae **20** coupled to the anterolateral papillary muscle **19**.

[0055] FIG. **7** is a process flow diagram showing an example of a process **700** for deploying a tether system as described herein, including one or more of the tether systems **400**, **500**, **600** described with reference to FIGS. **4**, **5** and **6**. In block **702**, the process can involve coupling a first anchor to a septal wall of a heart ventricle. In block **704**, the process can involve extending a portion of a tether within the heart ventricle from the first anchor to a portion of a ventricular heart wall of the heart ventricle. In block **706**, the process can involve positioning a second anchor over an externally oriented surface of the portion of the ventricular heart wall. In block **708**, the process can involve coupling the tether to the second anchor while the second anchor is over the externally oriented surface of the ventricular heart wall.

[0056] In some instances, coupling the first anchor to the septal wall can comprise embedding the first anchor within the septal wall. Extending the portion of the tether from the first anchor to the portion of the ventricular heart wall of the heart ventricle can comprise extending a portion of the tether through a portion of the septal wall. Alternatively, the first anchor may not be within the septal wall. In some instances, the first anchor can be disposed over a surface of the septal wall. For example, coupling the first anchor to the septal wall can comprise positioning the first anchor over a surface of the septal wall having an opposing orientation relative to that oriented toward the heart ventricle. The first anchor can be disposed in a different heart ventricle than that in which the tether extends. For example, the first anchor can be disposed over and be in contact with a surface of the septal wall oriented toward the different heart ventricle. Extending the portion of the tether from the first anchor to the portion of the ventricular heart wall of the heart ventricle can comprise extending a portion of the tether through an entire thickness of the septal wall, such as from the surface over which the first anchor is disposed to the surface of the septal wall oriented toward the heart ventricle within which the tether extends.

[0057] In some instances, coupling the first anchor to the septal wall can comprise coupling the first anchor to a basal portion of the septal wall. In some instances, coupling the first anchor to the septal wall can comprise coupling the first anchor to a mid-portion of the septal wall. In some instances, the second anchor can be disposed over an externally oriented surface of a mid-portion of the ventricular heart wall. For example, extending the portion of the tether from the first anchor to the portion of the ventricular heart wall of the heart ventricle can comprise extending the portion of the tether from the first anchor to a mid-portion of the ventricular heart wall. In some instances, extending the portion of the tether from the first anchor to the portion of the ventricular heart wall of the heart ventricle can comprise extending the portion of the tether from the first anchor coupled to a basal portion of the septal wall to a mid-portion of the ventricular heart wall. In some instances, extending the portion of the tether from the first anchor to the portion of the ventricular heart wall of the heart ventricle can comprise extending the portion of the tether from the first anchor coupled to a mid-portion of the septal wall to a mid-portion of the ventricular heart wall.

[0058] In some instances, extending the portion of the tether from the first anchor to the portion of the ventricular heart wall of the heart ventricle can comprise extending the portion of the tether directly from the first anchor to the second anchor. For example, the tether can extend between the first and second anchors without making any loops. Alternatively, the portion of the tether extending between the first and second anchors can form one or more loops. In some instances, extending the portion of the tether from the first anchor to the portion of the ventricular heart wall of the heart ventricle can comprise looping the tether around papillary muscles and/or chordae tendineae coupled to the papillary muscles. For example, a portion of the tether can form one or more loops around both papillary muscles and/or chordae tendineae coupled to the papillary muscles of the heart ventricle.

[0059] In some instances, the heart ventricle can be a left heart ventricle. Extending a portion of a tether within the heart ventricle from the first anchor to a portion of a ventricular heart wall of the heart ventricle can comprise extending a portion of a tether within the left heart ventricle from the first anchor to a portion of a ventricular heart wall of the left heart ventricle. In some instances, positioning a second anchor over an externally oriented surface of the portion of the ventricular heart wall can comprise positioning a second anchor over an externally oriented surface of the portion of the ventricular heart wall of the left heart ventricle. In some instances, coupling the tether to the second anchor while the second anchor is over the externally oriented surface of the ventricular heart wall can comprise coupling the tether to the second anchor while the second anchor is over the externally oriented surface of the ventricular heart wall of the left heart ventricle. Extending the portion of the tether from the first anchor to the portion of the ventricular heart wall of the heart ventricle can comprise looping the tether around both a posteromedial papillary muscle or chordae tendineae coupled to the posteromedial papillary muscle, and an anterolateral papillary muscle or chordae tendineae coupled to the anterolateral papillary muscle.

[0060] FIG. **8** shows a tether system **800** deployed in a heart **1** and configured to couple a papillary muscle of a heart ventricle, such as a left heart ventricle **3**, to a ventricular heart wall **17** of the heart ventricle **1**. The tether system **800** can comprise a first anchor **810** configured to be coupled to the papillary muscle of the heart ventricle. A second anchor **830** can be configured to be coupled to the ventricular heart wall **17** of the heart ventricle. A tether **850** can be configured to couple the first anchor **810** and the second anchor **830** to one another. For example, the tether **850** can comprise a portion, such as a medial portion **858**, configured to extend within the heart ventricle between the first anchor **810** and the second anchor **830**. A first distal portion **852** of the tether **850** can be coupled to the first anchor **810**. A second distal portion **854** of the tether **850** can be coupled to the second anchor **830**. The first anchor **810** can be configured to be disposed over a surface of the papillary muscle. The second anchor **830** can be configured to be positioned over an externally oriented surface the ventricular heart wall **17**. For example, at least a portion of the first distal portion **852** of the tether **850** can be configured to extend through a portion of the papillary muscle such that the first distal portion **852**, such as a first distal end **856**, can be coupled to the first anchor **810**. At least a portion of the second distal portion **854** of the tether **850** can be configured to extend through a portion of the ventricular heart wall **17** such that the second distal portion **854** can be coupled to the second anchor **830**.

[0061] In some instances, the second anchor **830** can be configured to be positioned over an externally oriented surface of an apical portion, including an apex, of the ventricular heart wall **17** of the heart ventricle. For example, at least a portion of the second distal portion **854** of the tether **850** can be configured to extend through the apical portion of the ventricular heart wall **17** such that the second distal portion **854** can be coupled to the second anchor **830**. In some instances, at least a portion of the second distal portion **854** of the tether **850** can be configured to extend through the apex of the ventricular heart wall **17** such that the second distal portion **854** can be coupled to the second anchor **830**. Coupling the papillary muscle to an apical portion of the ventricular heart wall **17** can be configured to pull the papillary muscle and/or a portion of the ventricular heart wall **17** adjacent to papillary muscle inward. For example, the papillary muscle and/or a portion of the ventricular heart wall **17** from which the papillary muscle extends can be pulled inward.

[0062] As described herein, in some instances, the heart ventricle can be the left ventricle **3**. The first anchor **810** can be configured to be coupled to a papillary muscle of the left ventricle **3**, including an anterolateral papillary muscle **19** of the left ventricle **3**. For example, the first anchor **810** can be disposed over a surface of the anterolateral papillary muscle **19**. The second anchor **830** can be configured to be disposed over an externally oriented surface of an apical portion, including an apex, of the left ventricular heart wall. At least a portion of the first distal portion **852** of the tether **850** can be configured to extend through the anterolateral papillary muscle **19** such that the first distal portion **852** can be coupled to the first anchor **810**. At least a portion of the second distal portion **854** of the tether **850** can be configured to extend through the apical portion of the left ventricular heart wall, such as the apex, such that the second distal portion **854** can be coupled to the second anchor **830**.

[0063] FIG. **9** is a process flow diagram showing an example of a process **900** for deploying a tether system as described herein, such as the tether system **800** described with reference to FIG. **8**. In block **902**, the process can involve coupling a first anchor to a papillary muscle of a heart ventricle. In block **904**, the process can involve extending a portion of a tether within the heart ventricle from the first anchor to an apical portion of a ventricular heart wall of the heart ventricle. In block **906**, the process can involve positioning a second anchor over an externally oriented surface of the apical portion of the ventricular heart wall. In block **908**, the process can involve coupling the tether to the second anchor while the second anchor is over the externally oriented surface of the apical portion of the ventricular heart wall.

[0064] In some instances, the heart ventricle can be the left heart ventricle. For example, coupling the first anchor to the papillary muscle can comprise coupling the first anchor to a papillary muscle of the left heart ventricle. In some instances, coupling the first anchor to the papillary muscle can comprise coupling the first anchor to an anterolateral papillary muscle of the left heart ventricle. For example, extending the portion of the tether within the heart ventricle from the first anchor to the apical portion of the ventricular heart wall can comprise extending the portion of the tether within the heart ventricle from the first anchor coupled to the anterolateral papillary muscle to the apical portion of the left heart ventricle.

[0065] In some instances, the papillary muscle can be coupled to an apex of the heart ventricle, such as the left heart ventricle. For example, extending the portion of the tether within the heart ventricle from the first anchor to the apical portion of the ventricular heart wall can comprise extending the portion of the tether within the heart ventricle from the first anchor to an apex of the ventricular heart wall. In some instances, extending the portion of the tether within the heart ventricle from the first anchor to the apical portion of the ventricular heart wall can comprise extending the portion of the tether within the heart ventricle from the first anchor coupled to the anterolateral papillary muscle of the left heart ventricle to an apex of the ventricular heart wall of the left heart ventricle.

[0066] FIG. **10** shows a tether system **1000** deployed in a heart **1** and configured to couple a first portion of a ventricular heart wall **17** and a second portion of the ventricular heart wall **17** of a heart ventricle to one another via a papillary muscle of the heart ventricle. The tether system **1000** can comprise a first anchor **1010** configured to be coupled to the first portion of the ventricular heart wall **17** and a second anchor **1030** configured to be coupled to the second portion of the ventricular heart wall **17**. A tether **1050** can be configured to couple the first anchor **1010** and the second anchor **1030** to one another. The tether **1050** can comprise a portion configured to form at least a partial loop around a papillary muscle of the heart ventricle or chordae tendineae coupled to the papillary muscle. For example, the tether **1050** can comprise a medial portion **1058** configured to extend within the heart ventricle between the first and second anchors **1010**, **1030**, and the portion of the tether **1050** configured to be disposed within the ventricle configured to form at least a partial loop around a papillary muscle of the heart ventricle or chordae tendineae coupled to the papillary muscle. The tether system **1000** can be configured to pull the portion of the ventricular heart wall inward and toward the papillary muscle.

[0067] The first anchor **1010** can be configured to be disposed over an externally oriented surface of a first portion of the ventricular heart wall **17**. The second anchor **1030** can be configured to be disposed over an externally oriented surface of a second portion of the ventricular heart wall **17**. The tether **1050** can be configured to extend between the first anchor **1010** and the second anchor **1030** while making at least a partial loop around the papillary muscle or chordae tendineae coupled to the papillary muscle. The tether **1050** can comprise a first distal portion **1052** configured to be coupled to the first anchor **1010** and a second distal portion **1054** of the tether **1050** can be coupled to the second anchor **1030**. For example, at least a portion of the first distal portion **1052** of the tether **1050** can be configured to extend through the first portion of the ventricular heart wall **17** such that the first distal portion **1052**, such as a first distal end **1056**, can be coupled to the first anchor **1010** disposed over the externally oriented surface of the first portion of the ventricular heart wall **17**. At least a portion of the second distal portion **1054** of the tether **1050** can be configured to extend through the second portion of the ventricular heart wall **17** such that the second distal portion **1054** can be coupled to the second anchor **1030** disposed over the externally oriented surface of the second portion of the ventricular heart wall **17**.

[0068] The first portion of the ventricular heart wall **17** can be above the second portion of the ventricular heart wall **17**. In some instances, at least one of the first portion and the second portion are in a mid-portion of the ventricular heart wall **17**. In some instances, at least one of the first portion and the second portion are in an apical portion of the ventricular heart wall **17**. For example, the first portion of the ventricular heart wall **17** can be in a mid-portion while the second portion of the ventricular heart wall **17** can be in the apical portion. In some instances, both the first and second portions of the ventricular heart wall **17** can be in the mid-portion. In some instances, both the first and second portions of the ventricular heart wall **17** can be in the apical portion.

[0069] In some instances, the heart ventricle can be the left ventricle **3**. The first anchor **1010** can be configured to be disposed over an externally oriented surface of a first portion of the ventricular heart wall **17** of the left ventricle **3**. The second anchor **1030** can be configured to be disposed over an externally oriented surface of a second portion of the ventricular heart wall **17** of the left ventricle **3**. The tether **1050** can be configured to extend between the first anchor **1010** and the second anchor **1030** while making at least a partial loop around a papillary muscle or chordae tendineae coupled to the papillary muscle of the left ventricle **3**, including the posteromedial papillary muscle **21** of the left ventricle **3**. For example, a portion of the tether **1050** can form a partial loop around the posteromedial papillary muscle **21** or chordae tendineae **22** coupled to the posteromedial papillary muscle **21** of the left ventricle **3**.

[0070] In some instances, a pad **1040** can be deployed over and/or in contact with the first portion of the ventricular heart wall **17** to facilitate maintaining the first anchor **1010** at the desired location. For example, a first surface **1042** of the pad **1040** can be configured to be oriented toward and/or be in contact with the ventricular heart wall **17** and a second surface **1044** of the pad **1040** can be configured to be oriented away from the ventricular heart wall **17**. The first anchor **1010** can be disposed over and/or in contact with the second surface **1044**. The first distal portion **1052** of the tether **1050** can extend through the pad **1040** to couple to the first anchor **1010**.

[0071] FIG. **11** is a process flow diagram of an example of a process **1100** for deploying a tether system as described herein, including the tether system **1000** described with reference to FIG. **10**. In block **1102**, the process can involve coupling a first anchor to a first portion of a ventricular heart wall of a heart ventricle. In block **1104**, the process can involve extending a portion of a tether within the heart ventricle from the first anchor to a second portion of the ventricular heart wall of the heart ventricle and forming at least a partial loop, using the tether, around a papillary muscle of the heart ventricle or chordae tendineae coupled to the papillary muscle. In block **1106**, the process can involve positioning a second anchor over an externally oriented surface of the second portion of the ventricular heart wall. In block **1108**, the process can involve coupling the tether to the second anchor while the second anchor is over the externally oriented surface of the ventricular heart wall, such as while the second anchor is over the second portion of the ventricular heart wall.

[0072] The first portion of the ventricular heart wall can be at a higher position on the ventricular wall than the second portion of the ventricular heart wall. For example, extending the portion of the tether within the heart ventricle from the first anchor to the second portion of the ventricular heart wall can comprise extending the portion of the tether within the heart ventricle from the first anchor to a second portion of the ventricular heart wall that is lower on the ventricular heart wall than the first portion.

[0073] In some instances, both the first portion and the second portion of the ventricular heart wall can be at respective positions on a mid-portion of the ventricular heart wall. For example, coupling the first anchor to the first portion of the ventricular heart wall can comprise coupling the first anchor to a first mid-portion of the ventricular heart wall. In some instances, extending the portion of the tether configured to be within the heart ventricle from the first anchor to the second portion of the ventricular heart wall can comprise extending the portion of the tether within the heart ventricle from the first anchor to a second mid-portion of the ventricular heart wall. As described herein, the first portion of the ventricular heart wall can be at a higher position on the ventricular wall than the second portion of the ventricular heart wall. In some instances, extending the portion of the tether configured to be within the heart ventricle from the first anchor to the second portion of the ventricular heart wall can comprise extending the portion of the tether within the heart ventricle from the first anchor to a second mid-portion of the ventricular heart wall that is at a position lower than the first portion of the ventricular heart wall.

[0074] In some instances, the first portion of the ventricular heart wall can be on a mid-portion of the ventricular heart wall while the second portion of the ventricular heart wall can be on an apical portion of the ventricular heart wall. For example, coupling the first anchor to the first portion of the ventricular heart wall can comprise coupling the first anchor to a mid-portion of the ventricular heart wall. Extending the portion of the tether configured to be within the heart ventricle from the first anchor to the second portion of the ventricular heart wall can comprises extending the portion of the tether within the heart ventricle from the first anchor to an apical portion of the ventricular heart wall.

[0075] In some instances, both the first portion and the second portion of the ventricular heart wall can be at respective positions on an apical portion of the ventricular heart wall. For example, coupling the first anchor to the first portion of the ventricular heart wall can comprise coupling the first anchor to a first apical portion of the ventricular heart wall. In some instances, extending the portion of the tether configured to be within the heart ventricle from the first anchor to the second portion of the ventricular heart wall can comprise extending the portion of the tether within the heart ventricle from the first anchor to a second apical portion of the ventricular heart wall. As described herein, the first portion of the ventricular heart wall can be at a higher position on the ventricular wall than the second portion of the ventricular heart wall. In some instances, extending the portion of the tether configured to be within the heart ventricle from the first anchor to the second portion of the ventricular heart wall can comprise extending the portion of the tether within the heart ventricle from the first anchor to a second apical portion of the ventricular heart wall that is at a position lower than the first portion of the ventricular heart wall.

[0076] In some instances, forming at least a partial loop, using the tether, around the papillary muscle can comprise forming at least a partial loop, using the tether, around a posteromedial papillary muscle or chordae tendineae coupled to the posteromedial papillary muscle. For example, a partial loop can be formed around the posteromedial papillary muscle or chordae tendineae coupled to the posteromedial papillary muscle.

[0077] It will be understood that the tether systems **400**, **500**, **600**, **800**, **1000** described with reference to FIGS. **4**, **5**, **6**, **8** and **10** can have one or more features of the tether systems **100**, **200** described with reference to FIGS. **1** and **2**. For example, each of the second anchors **430**, **530**, **630**, **830**, **1030** can comprise a first surface **432**, **532**, **632**, **832**, **1032** configured to be oriented toward the ventricular heart wall and a second surface **434**, **534**, **634**, **834**,**1034** configured to be oriented away from the ventricular heart wall. The first surface **432**, **532**, **632**, **832**, **1032** can be configured to be over and in contact with the ventricular heart wall. In some instances, the second distal portion **454**, **554**, **654**, **854**, **1054** of the tether **450**, **550**, **650**, **850**, **1050** can be configured to extend through the respective ventricular heart wall and the second anchor **430**, **530**, **630**, **830**, **1030** such that a knot **470**, **570**, **670**, **870**, **1070** can be formed over the second surface **434**, **534**, **634**, **834**,**1034** of the respective second anchor **430**, **530**, **630**, **830**, **1030** to couple the tether **450**, **550**, **650**, **850**, **1050** to the second anchor **430**, **530**, **630**, **830**, **1030**.

[0078] In some instances, the first anchor **410**, **510**, **610**, **810**, **1010** and the tether **450**, **550**, **650**,

850, **1050** can be an integral piece. In some instances, the first anchor **410**, **510**, **610**, **810**, **1010** and the tether **450**, **550**, **650**, **850**, **1050** can be formed using the same material. The first anchor **410**, **510**, **610**, **810**, **1010** and the tether **450**, **550**, **650**, **850**, **1050** can each comprise a respective portion of a surgical cord and/or suture. For example, a portion of a suture **402**, **502**, **602**, **802**, **1002** can form the first anchor **410**, **510**, **610**, **810**, **1010**. Another portion of the suture **402**, **502**, **602**, **802**, **1002** can form the tether **450**, **550**, **650**, **850**, **1050**. As described in further detail herein, in some instances, the first anchor **410**, **510**, **610**, **810**, **1010** can comprise a bulky-knot. For example, a portion of the suture **402**, **502**, **602**, **802**, **1002** can be deployed through the portion of the septal wall, papillary muscle or ventricular heart wall to form the first anchor **410**, **510**, **610**, **810**, **1010**. Another portion of the suture **402**, **502**, **602**, **802**, **1002** can form the tether **450**, **550**, **650**, **850**, **1050**, for example extending through the septal wall, papillary muscle or ventricular heart wall, the ventricle, and another portion of the ventricular wall, such that suture **402**, **502**, **602**, **802**, **1002** can be coupled to the second anchor **430**, **530**, **630**, **830**, **1030**.

[0079] In some instances, each of the tethers **450**, **550**, **650**, **850**, **1050** can comprise respective tether tails **460**, **560**, **660**, **860**, **1060**. For example, the tethers **450**, **550**, **650**, **850**, **1050** can each comprise a pair of tether tails **460***a*, **460***b*, **560***a*, **560***b*, **660***a*, **66***ob*, **860***a*, **860***b*, **1060***a*, **1060***b*, each comprising a respective first distal portion **462***a*, **462***b*, **562***a*, **562***b*, **662***a*, **662***b*, **862***a*, **862***b*, **1062***a*, **1062***b* configured to extend through the portion of the septal wall, papillary muscle or ventricular heart wall and a second distal portion **464***a*, **464***b*, **564***a*, **564***b*, **664***a*, **664***b*, **864***a*, **864***b*, **1064***a*, **1064***b* configured to extend through another portion of the ventricular heart wall and couple to the respective second anchors **430**, **530**, **630**, **830**, **1030**. For example, the first distal portion **462***a*, **462***b*, **562***a*, **562***b*, **662***a*, **662***b*, **862***a*, **862***b*, **1062***a*, **1062***b*, including the first distal end **466***a*, **466***b*, **566***a*, **566***b*, **666***a*, **666***b*, **866***a*, **866***b*, **1066***a*, **1066***b*, can be configured to the first anchor **410**, **510**, **610**, **810**, **1010**. A medial portion **468***a*, **468***b*, **568***a*, **568***b*, **668***a*, **668***b*, **868***a*, **868***b*, **1068***a*, **1068***b* can be configured to extend through the ventricle. The second distal portions **464***a*, **464***b*, **564***a*, **564***b*, **664***a*, **664***b*, **864***a*, **864***b*, **1064***a*, **1064***b* can be configured to extend through the respective second anchors **430**, **530**, **630**, **830**, **1030** and form the knot **470**, **570**, **670**, **870**, **1070** for coupling the tethers **450**, **550**, **650**, **850**, **1050** to the second anchors **430**, **530**, **630**, **830**, **1030**.

[0080] In some instances, the tether systems **400**, **500**, **600**, **800**, **1000** can optionally comprise an elongate tube **480**, **580**, **680**, **880**, **1080** comprising a lumen **482**, **582**, **682**, **882**, **1082** configured to receive at least a portion of the tether **450**, **550**, **650**, **850**, **1050** extending within the heart ventricle. The elongate tube **480**, **580**, **680**, **880**, **1080** can be configured to prevent or reduce damage to heart tissue due to contact between the tether **450**, **550**, **650**, **850**, **1050** and the heart tissue. For example, a process for deploying the tether system **400**, **500**, **600**, **800**, **1000** can comprise advancing the elongate tube **480**, **580**, **680**, **880**, **1080** along at least a portion of a respective tether **450**, **550**, **650**, **850**, **1050** within the heart ventricle. In some instances, the second distal portion **454**, **554**, **654**, **854**, **1054** the respective tether **450**, **550**, **650**, **850**, **1050** can be positioned into the lumen **482**, **582**, **682**, **882**, **1082** of the elongate tube **480**, **580**, **680**, **880**, **1080** such that the elongate tube elongate tube **480**, **580**, **680**, **880**, **1080** can be advanced along the respective tether **450**, **550**, **650**, **850**, **1050**.

[0081] FIG. **12** is a perspective view of a tether delivery system **1200** in accordance with one or more examples. The tether delivery system **1200** may be used in the deployment of one or more tether systems as described herein. For example, the tether delivery system **1200** can be used in the deployment of the tether system **100** described with reference to FIG. **1**. FIGS. **13**A-**13**E describe an example of a process for using the tether delivery system **1200** to deliver the first anchor **110** and tether **150** of the tether system **100** to a heart valve annulus. The tether delivery system **1200** may be utilized to deliver and anchor tissue anchors, such as suture-knot-type tissue anchors, to one or more portions of the heart, such as a heart valve annulus, septal wall, papillary muscle, and/or ventricular heart wall. Such procedures may be implemented on a beating heart.

[0082] The tether delivery system **1200** can include an elongate rigid tube **1210** forming at least one internal working lumen. Although described in certain examples and/or contexts as comprising an elongate rigid tube, it should be understood that tubes, shafts, lumens, conduits, and the like disclosed herein may be either rigid, at least partially rigid, at least flexible, and/or at least partially flexible. Therefore, any such component described herein, whether or not referred to as rigid herein, should be interpreted as possibly being at least partially flexible. In alternative instances, an articulating and/or steerable elongate tube can be used to facilitate navigation within the patient anatomy to facilitate deployment of the tether systems.

[0083] In addition to the elongate rigid tube **1210**, the tether delivery system **1200** may include a plunger feature **1240**, which may be used or actuated to manually deploy the tether. The tether delivery system **1200** may further include a plunger lock mechanism **1245**, which may serve as a safety lock that locks the tether delivery system **1200** until ready for use or deployment of a tether as described herein. The plunger feature **1240** may have associated therewith a suture-release mechanism, which may be configured to lock in relative position a pair of tether tails **160***a*, **16***ob* associated with a pre-formed suture knot sutureform anchor (not shown) to be deployed. The tether delivery system **1200** may further comprise a flush port **1250**, which may be used to de-air the lumen of the elongate rigid tube **1210**. For example, heparinized saline flush, or the like, may be connected to the flush port **1250** using a female Luer fitting to de-air the tether delivery system **1200**. The term “lumen” is used herein according to its broad and ordinary meaning, and may refer to a physical structure forming a cavity, void, pathway, or other channel, such as an at least partially rigid elongate tubular structure, or may refer to a cavity, void, pathway, or other channel, itself, that occupies a space within an elongate structure (e.g., a tubular structure). Therefore, with respect to an elongate tubular structure, such as a shaft, tube, or the like, the term “lumen” may refer to the elongate tubular structure and/or to the channel or space within the elongate tubular structure.

[0084] The lumen of the elongate rigid tube **1210** may house a needle (not shown) that is wrapped at least in part with a pre-formed suture knot sutureform anchor, as described in detail herein. In some instances, the elongate rigid tube **1210** presents a relatively low profile. For example, the elongate rigid tube **1210** may have a diameter of about 3 millimeters (mm) or less (e.g., about 9 French (Fr) or less). The elongate rigid tube **1210** can be associated with an atraumatic tip **1214** feature. The atraumatic tip **1214** can be an echogenic positioner component, which may be used for deployment and/or positioning of the suture-type tissue anchor. The atraumatic tip **1214**, disposed at the distal end of the elongate rigid tube **1210**, may be configured to have deployed therefrom a wrapped pre-formed suture knot sutureform anchor. For example, deployment of the wrapped pre-formed suture knot sutureform anchor can comprise advancing the wrapped pre-formed suture knot sutureform anchor through a distal end opening **1212** of the elongate rigid tube **1210** while the atraumatic tip **1214** is in contact with target tissue, as described herein

[0085] The atraumatic tip **1214** may be referred to as an “end effector.” In addition to a pre-formed suture knot sutureform anchor and associated needle, the elongate rigid tube **1210** may house an elongated knot pusher tube (not shown; also referred to herein as a “pusher”), which may be actuated using the plunger **1240** in some examples. As described in further detail below, the atraumatic tip **1214** can provide a surface against which the target heart surface portion may be held to facilitate deployment of an anchor through the target heart surface portion.

[0086] The tether delivery system **1200** may be used to deliver a “bulky-knot” type tissue anchor to a target heart location, as described in greater detail below. The atraumatic tip **1214** can be coupled to the distal end portion of the elongate rigid tube **1210**, wherein the proximal end portion of the elongate rigid tube **1210** may be coupled to a handle portion **1220** of the tether delivery system **1200**, as shown. The atraumatic tip **1214** (e.g., end effector), can be placed in contact with a surface of the target heart location to facilitate deployment of the “bulky-knot” anchor. Generally, the elongate pusher (not shown) may be movably disposed within a lumen of the elongate rigid tube **1210** and coupled to a pusher hub (not shown) that is movably disposed within the handle **1220** and releasably coupled to the plunger **1240**. A needle (not shown) carrying a pre-formed suture knot sutureform anchor can be movably disposed within a lumen of the pusher and coupled to a needle hub (not shown) that is also coupled to the plunger **1240**. The “bulky-knot” anchor can be formed using the pre-formed suture knot sutureform anchor carried on the needle. The plunger **1240** can be used to actuate or move the needle and the pusher during deployment of the “bulky-knot” anchor (see, e.g., FIGS. **13**C and **13**D). The plunger **1240** can be movably disposed at least partially within the handle **1220**. For example, the handle **1220** may define a lumen in which the plunger **1240** can be moved. During operation, the pusher may also move within the lumen of the handle **1220**. The plunger lock **1245** can be used to prevent the plunger **1240** from moving within the handle **1220** during storage and prior to performing a procedure to deploy a tissue anchor.

[0087] The needle may have the pre-formed suture knot sutureform anchor disposed about a distal portion thereof while maintained in the elongate rigid tube **1210**. For example, the pre-formed suture knot sutureform anchor may be formed of one or more sutures configured in a coiled sutureform (see FIG. **13**C) having a plurality of winds/turns around the needle over a portion of the needle that is associated with a longitudinal slot in the needle that runs from the distal end thereof. Although the term “sutureform” is used herein, it should be understood that such components/forms may comprise suture, wire, or any other elongate material wrapped or formed in a desired configuration. The coiled sutureform can be provided or shipped disposed around the needle. In some examples, two tether tails extend from the coiled sutureform. FIG. **12** shows the tether tails **160***a*, **160***b* of the tether **150** extending through the lumen of the needle and/or through a passageway of the plunger **1240** and may exit the plunger **1240** at a proximal end portion thereof. The coiled sutureform may advantageously be configured to be formed into a suture-type tissue anchor (referred to herein as a “bulky-knot”) in connection with an anchor-deployment procedure, as described in more detail below. The coiled sutureform can be configurable to a knot/deployed configuration by approximating opposite ends of the coiled portion thereof towards each other to form one or more loops.

[0088] The tether delivery system **1200** can further include a suture/tether catch mechanism (not shown) coupled to the plunger **1240** at a proximal end of the tether delivery system **1200**, which may be configured to releasably hold or secure the tether **150**, such as the tether tails **160***a*, **16***ob*, extending through the tether delivery system **1200** during delivery of a tissue anchor as described herein. The suture catch can be used to hold the tether **150** with a friction fit or with a clamping force and can have a lock that can be released after the tissue anchor has been deployed/formed into a bulky-knot, as described herein.

[0089] As described herein, the tether delivery system **1200** can be used in beating heart repair procedures. In some examples, the elongate rigid tube **1210** of the tether delivery system **1200** can be configured to extend and contract with the beating of the heart. During systolic contraction, the median axis of the heart generally shortens. For example, the distance from the apex of the heart to the valve leaflets can vary by about 1 centimeter (cm) to about 2 centimeters (cm) with each heartbeat in some patients. In some examples, the length of the elongate rigid tube **1210** that protrudes from the handle **1220** can change with the length of the median axis of the heart. That is, distal end of the elongate rigid tube **1210** can be configured to be floating such that the shaft can extend and retract with the beat of the heart so as to maintain contact with the target mitral valve leaflet.

[0090] Implementation of a repair procedure utilizing the tether delivery system **1200** can be performed in conjunction with certain imaging technology designed to provide visibility of the elongate rigid tube **1210** according to a certain imaging modality, such as echo imaging. Advancement of the tether delivery system **1200** may be performed in conjunction with echo imaging, direct visualization (e.g., direct transblood visualization), and/or any other suitable remote visualization technique/modality. With respect to cardiac procedures, for example, the tether delivery system **1200** may be advanced in conjunction with transesophageal (TEE) guidance and/or intracardiac echocardiography (ICE) guidance to facilitate and to direct the movement and proper positioning of the device for contacting the appropriate target cardiac region and/or target cardiac tissue (e.g., a valve leaflet, a valve annulus, or any other suitable cardiac tissue). Typical procedures that can be implemented using echo guidance are set forth in Suematsu, Y., *J. Thorac. Cardiovasc. Surg.* 2005; 130:1348-56 ("Suematsu"), the entire disclosure of which is incorporated herein by reference.

[0091] FIGS. **13**A through **13**E show various steps for delivering the tether **150** using the tether delivery system **1200** described with reference to FIG. **12**. FIG. **13**A is a cutaway view of a portion of the heart, and the tether delivery system **1200** disposed at least partially within a chamber of a heart in accordance with one or more examples. According to some implementations of repair procedures, an incision into the heart wall of the appropriate ventricle of the heart is made. FIG. **13**A shows the tether delivery system **1200** at least partially positioned within the left ventricle **3**. For instance, an introducer port device **1300** containing one or more fluid-retention valves to prevent blood loss from and/or air entry into the left ventricle **3**, may be inserted into the site of entry. Once the introducer port device **1300** is inside the left ventricle **3**, the elongate rigid tube **1210** may be advanced through the lumen **1320** of the introducer port device **1300**. In some examples, a sheath may be inserted through the introducer port device **1300**, through which one or more other instruments are advanced. For instance, an endoscope may first be advanced into the left ventricle **3** to visualize the ventricle **3**, the mitral valve **6**, and/or the sub-valvular apparatus. By use of an appropriate endoscope, a careful analysis of the malfunctioning mitral valve **6** may be performed.

[0092] One or more incisions may be made proximate to the thoracic cavity to provide a surgical field of access. The total number and length of the incisions to be made depend on the number and types of the instruments to be used as well as the procedure(s) to be performed. The incision(s) may advantageously be made in such a manner as to be minimally invasive. As referred to herein, the term "minimally invasive" means in a manner by which an interior organ or tissue may be accessed with relatively little damage being done to the anatomical structure through which entry is sought. For example, a minimally invasive procedure may involve accessing a body cavity by a small incision/opening of, for example, about 5 centimeters (cm) or less made in the skin of the body. The incision may be vertical, horizontal, or slightly curved. If the incision is located along one or more ribs, it may advantageously follow the outline of the rib. The opening may advantageously extend deep enough to allow access to the thoracic cavity between the ribs or under the sternum and is preferably set close to the rib cage and/or diaphragm, dependent on the entry point chosen.

[0093] In one example method, the heart may be accessed through one or more openings made by one or more small incision in a portion of the body proximal to the thoracic cavity, such as between one or more of the ribs of the rib cage of a patient, proximate to the xyphoid appendage, or via the abdomen and diaphragm. Access to the thoracic cavity may be sought to allow the insertion and use of one or more thorascopic instruments, while access to the abdomen may be sought to allow the insertion and use of one or more laparoscopic instruments. Insertion of one or more visualizing instruments may then be followed by transdiaphragmatic access to the heart. Additionally, access to the heart may be gained by direct puncture (e.g., via an appropriately sized needle, for instance an 18-gauge needle) of the heart from the xyphoid region. Accordingly, the one or more incisions should be made in such a manner as to provide an appropriate surgical field and access site to the heart in the least invasive manner possible. Access may also be achieved using percutaneous methods, further reducing the invasiveness of the procedure. See, e.g., "Full-Spectrum Cardiac Surgery Through a Minimal Incision Mini-Sternotomy (Lower Half) Technique," Doty et al., *Annals of Thoracic Surgery* 1998; 65(2): 573-7 and "Transxiphoid Approach Without Median Sternotomy for the Repair of Atrial Septal Defects," Barbero-Marcial et al., *Annals of Thoracic Surgery* 1998; 65(3): 771-4, the entire disclosures of each, which are incorporated herein by reference for all purposes.

[0094] Generally, the elongate rigid tube **1210** of the tether delivery system **1200** may be slowly advanced into the introducer **1300** until the atraumatic tip **1214** is distal to the introducer **1300** and/or cannula tip **1230** thereof and entered the left ventricle **3**. For example, the introducer **1300** can comprise a hub **1302** and an elongate shaft **1304** extending distally from the hub **1302**. The shaft **1304** can comprise at least a portion configured to be disposed within the left ventricle **3**, such as through an incision and/or opening **40** formed in the ventricular heart wall **17** of the left ventricle **3**. The elongate tube **1210** of the tether delivery system **1200** can be advanced until the atraumatic tip **1214** is positioned distally of the shaft **1304**. In so doing, it may be desirable to advance the elongate rigid tube **1210** within the left ventricle **3** in such a way as to avoid traversing areas populated by papillary muscles and/or associated chordae tendineae to avoid entanglement therewith. In order to facilitate or ensure avoidance of such anatomy, imaging technology may advantageously be implemented to provide at least partial visibility of the elongate rigid tube **1210** within the left ventricle **3**, as well as certain anatomical features within the ventricle. In some implementations, hybrid imaging technologies may be used, wherein echo imaging is used in combination with a different imaging modality. Multi-imaging modalities may provide improved visibility of anatomical and/or delivery system components.

[0095] FIG. **13**B shows a close-up view of the elongate rigid tube **1210** of the tether delivery system **1200** inserted into the left ventricle **3** and approximated to a target portion of the annulus adjacent to the mitral valve leaflet **64**, which may be a posterior leaflet of the mitral valve **6**, in connection with a repair procedure in accordance with one or more examples of the present disclosure. The elongate rigid tube **1210** can be configured to deliver a tissue anchor (not shown; see, e.g., FIGS. **13**C through **13**E), such as a bulky-knot, to the portion of the annulus adjacent to the valve leaflet **64**. It will be understood that the elongate rigid tube **1210** can also deliver a tissue anchor to the anterior mitral valve leaflet. Although the description of FIGS. **13**A through **13**E below is presented in the context of a mitral valve, it should be understood that the principles disclosed herein are applicable to other valves or biological tissues, such as a tricuspid valve.

[0096] With reference to FIGS. **13**A through **13**E, the elongate rigid tube **1210** can comprise one or more elongate lumens configured to allow delivery of the anchor **110** to the annulus of the mitral valve **6**. The elongate rigid tube **1210** can be configured to facilitate performance of one or more functions, such as grasping, suctioning, irrigating, cutting, suturing, or otherwise engaging a valve leaflet. The distal end, or atraumatic tip, **1214** of the elongate rigid tube **1210** can be configured to contact the portion of the annulus adjacent to the mitral valve leaflet **64** without substantially damaging the annulus. For example, during a repair procedure, a handle (e.g., handle **1220**) coupled to the elongate rigid tube **1210** can be manipulated in such a manner so that the annulus is contacted with the functional distal portion of the elongate rigid tube **1210** and a repair effectuated.

[0097] Echo imaging guidance, such as transesophageal echocardiogram (TEE) (**2**D and/or **3**D), transthoracic echocardiogram (TTE), and/or intracardiac echo (ICE), may be used to assist in the advancement and desired positioning of the elongate rigid tube **1210** within the left ventricle **3**. The atraumatic tip **1214** of the elongate rigid tube **1210** can contact a proximal surface (e.g., underside surface with respect to the illustrated orientation of FIGS. **13**B through **13**E) of the portion of the annulus adjacent to the mitral valve leaflet **64**, without or substantially without damaging the annulus. For example, the atraumatic tip **1214** can have a relatively blunt form or configuration. The atraumatic tip **1214** can be configured to maintain contact with the proximal side of the annulus as the heart beats to facilitate reliable delivery of the anchor **110** to the target site on the annulus.

[0098] In some examples, one or more perforation devices **1230** (e.g., needle(s)) can be delivered through a working lumen (not shown) of the elongate rigid tube **1210** to the portion of the mitral valve annulus adjacent to the valve leaflet **64** to puncture the portion of the annulus and project a sutureform **190** including a plurality of winds of suture about a distal portion of a needle **1230** into the left atrium **2** (see FIG. **13**C), wherein the sutureform **190** is deployed to form the bulky-knot anchor **110** shown in FIGS. **13**D and **13**E. As described herein, in some instances, the first anchor **110** and the tether **150** can be an integral piece formed using a suture **102**. For example a portion of the suture **102** can form the sutureform **190**. Another portion of the suture **102** can form the tether **150**, including the tether tails **160***a*, **16***ob* of the tether **150**. For example, as shown in FIG. **13**C, the needle **1230** can comprise a slot and can be deployed from the distal end of the elongate rigid tube **1210**, thereby puncturing the annulus and projecting into the left atrium **2**, wherein the slotted needle **1230** is wrapped with a sutureform **190** in a particular configuration (see PCT Application No. PCT/US2012/043761, published as WO 2013/003228 A1, for further detail regarding example suture wrapping configurations and needles for use in suture anchor deployment devices and methods, the entirety of which is incorporated herein by reference). In some instances, a pusher or hollow guide wire (not shown) is provided on or at least partially around the needle **1230** within the elongate rigid tube **1210**, such that the needle may be withdrawn, leaving the pusher and wound sutureform **190**. When a withdrawal force is applied to the sutureform **190** using the pusher, the sutureform **190** may form a bulky-knot-type anchor (e.g., the anchor **110**), after which the pusher may be withdrawn, leaving the bulky-knot-type anchor to anchor the tether tails **160***a*, **160***b* to the annulus.

[0099] FIG. **13**B shows the elongate rigid tube **1210** of the tether delivery system **1200** positioned on the mitral valve annulus adjacent to the mitral valve leaflet **64**. For example, the target site of the annulus may be slowly approached from the ventricle side thereof by advancing the distal end of elongate rigid tube **1210** along or near to the posterior wall of the left ventricle **3** without contacting the ventricle wall. Successful targeting and contacting of the target location on the annulus can depend at least in part on accurate visualization of the elongate rigid tube **1210** and/or atraumatic tip **1214** throughout the process of advancing the atraumatic tip **1214** to the target site. Generally, echocardiographic equipment may be used to provide the necessary or desired intra-operative visualization of the elongate rigid tube **1210** and/or atraumatic tip **1214**.

[0100] Once the atraumatic tip **1214** is positioned in the desired position, the distal end of the elongate rigid tube **1210** and the atraumatic tip **1214** may be used to drape, or “tent,” the annulus to better secure the atraumatic tip **1214** in the desired position, as shown in FIG. **13**B. Draping/tenting may advantageously facilitate contact of the atraumatic tip **1214** with the annulus throughout one or more cardiac cycles, to thereby provide more secure or proper deployment of leaflet anchor(s). Navigation of the atraumatic tip **1214** to the desired location on the underside of the mitral valve annulus may be assisted using echo imaging, which may be relied upon to confirm correct positioning of the atraumatic tip **1214** prior to anchor/knot deployment.

[0101] With the elongate rigid tube **1210** positioned against the mitral valve annulus adjacent to the mitral valve leaflet **64**, the plunger **1240** of the tether delivery system **1200** can be actuated to move the needle **1230** and a pusher disposed within the elongate rigid tube **1210**, such that the coiled sutureform portion **190** of the suture anchor slides off the needle **1230**, as shown in FIG. **13**C. As the plunger **1240** is actuated, a distal piercing portion of the needle **1230** punctures the annulus and forms an opening in the annulus. FIG. **13**C shows a close-up view of the distal portion of the elongate rigid tube **1210** showing the needle **1230** and tissue anchor sutureform **190** projected therefrom through the annulus in accordance with one or more examples. In some examples, the needle **1230** is projected a distance of between about 0.2 to about 0.3 inches (about 5 to about 8 millimeters (mm)), or less, distally beyond the distal end of the elongate rigid tube **1210** (e.g., beyond the atraumatic tip **1214**). In some examples, the needle **1230** is projected a distance of between about 0.15 to about 0.4 inches (about 4 to about 10 millimeters (mm)). In some examples, the needle **1230** is projected a distance of about 1 inch (about 25 millimeters (mm)), or greater. In some examples, the needle **1230** extends until the distal tip of the needle and the entire coiled sutureform **190** extend through the annulus. While the needle **1230** and sutureform **190** are projected into the side of the annulus oriented toward the left atrium **2**, the elongate rigid tube **1210** and atraumatic tip **1214** advantageously remain entirely on the side of the annulus oriented toward the left ventricle **3**.

[0102] As the pusher (not shown) within the tissue anchor delivery device elongate rigid tube **1210** is moved distally, a distal end of the pusher advantageously moves or pushes the distal coiled sutureform **190** (e.g., pre-deployment coiled portion of the suture anchor) over the distal end of the needle **1230** and further within the left atrium **2** of the heart on a distal side of the mitral valve annulus adjacent to the mitral valve leaflet **64**, such that the sutureform **190** extends distally beyond a distal end of the needle **1230**. For example, in some examples, at least half a length of the sutureform **190** extends beyond the distal end of the needle **1230**. In some examples, at least three quarters of the length of the sutureform **190** extends beyond the distal end of the needle **1230**. In some examples, the entire coiled sutureform **190** extends beyond the distal end of the needle **1230**.

[0103] After the sutureform **190** has been pushed off the needle **1230**, pulling one or more of the tether tail(s) **160***a*, **160***b* (e.g., strands extending from the coiled portion of the tether) associated with the tissue anchor **110** proximally can cause the sutureform **190** to form the anchor **110** (e.g., a bulky-knot anchor), as shown in FIG. **13**D, which provides a close-up view of the formed anchor **110** on the side of the annulus oriented toward the left atrium **2**. For example, the bulky-knot suture anchor may be formed by approximating opposite ends of the coils of the sutureform **190** (see FIG. **13**C) towards each other to form one or more loops. After the sutureform **190** has been formed into the bulky-knot, the tether delivery system **1200** can be withdrawn proximally, leaving the anchor **110** disposed on the distal atrial side of the annulus.

[0104] FIG. **13**E shows a cutaway view of the heart with the deployed anchor **110** and the elongate rigid tube **1210** being withdrawn from the mitral valve annulus after the anchor **110** has been formed. The tether tails **160***a*, **160***b* extend proximally from the anchor **110**. In some examples, the two tether tails **160***a*, **16***ob* may extend from the proximal side of the annulus, such as the side of the annulus oriented toward the left ventricle **3**. The tether tails **160***a*, **16***ob* can extend out of the heart **1** through the opening **40** formed in the ventricular wall. For example, the elongate rigid tube **1210** can be slid/withdrawn over the tether tails **160***a*, **16***ob*. The tether tails **160***a*, **16***ob* can advantageously be tensioned to facilitate desired reshaping of the left ventricle. The tether tails **160***a*, **160***b* coupled to the anchor **110** may be secured at the desired tension using a surgical pad as described herein. One or more knots (e.g., a knot stack) or other suture fixation mechanism(s) or device(s) may be implemented to hold the tether tails **160***a*, **16***ob* at the desired tension and to the surgical pad. With the tether tails **160***a*, **16***ob* fixed to the ventricle wall, a ventricular portion of the tether tails **160***a*, **16***ob* may advantageously function to tether the mitral valve annulus in a desired manner and at a desired tension, facilitating reshaping of left ventricular wall.

ADDITIONAL DESCRIPTION OF EXAMPLES

[0105] Provided below is a list of examples, each of which may include aspects of any of the other examples disclosed herein. Furthermore, aspects of any example described above may be implemented in any of the numbered examples provided below. [0106] Example 1: A tether system comprising a first anchor configured to be coupled to an annulus of a heart valve that is between a heart atrium and a heart ventricle, a second anchor configured to be positioned over an externally oriented surface of a portion of a ventricular heart wall of the heart ventricle between an apex and a basal portion of the ventricular heart wall, and a tether configured to couple the first anchor and the second anchor to one another. [0107] Example 2: The system of any of the examples described herein, in particular example 1, wherein the portion of the ventricular heart wall is in a mid-portion of the ventricular heart wall. [0108] Example 3: The system of any of the examples described herein, in particular example 1, wherein the portion of the ventricular heart wall is between an apical portion and a mid-portion of the ventricular heart wall. [0109] Example 4: The system of any of the examples described herein, in particular example 1, wherein the portion of the ventricular heart wall is adjacent to a portion of the ventricular heart wall from which a papillary muscle extends. [0110] Example 5: The system of any of the examples described herein, in particular example 4, wherein the portion of the ventricular heart wall is adjacent to a portion of the ventricular heart wall from which an anterolateral papillary muscle extends. [0111] Example 6: The system of any of the examples described herein, in particular examples 1 to 5, wherein the first anchor is configured to be coupled to an annulus of a mitral valve and the heart ventricle is a left ventricle. [0112] Example 7: The system of any of the examples described herein, in particular example 6, wherein the first anchor is configured to be coupled to a portion of the annulus adjacent to a posterior leaflet of the mitral valve. [0113] Example 8: The system of any of the examples described herein, in particular example 6, wherein the first anchor is configured to be coupled to a portion of the annulus adjacent to an anterior leaflet of the mitral valve. [0114] Example 9: The system of any of the examples described herein, in particular examples 1 to 8, further comprising an elongate tube configured to receive at least a portion of the tether extending within the heart ventricle. [0115] Example 10: The system of any of the examples described herein, in particular examples 1 to 9, wherein the tether comprises a suture. [0116] Example 11: The system of any of the examples described herein, in particular example 10, wherein the first anchor comprises at least a portion formed using the suture. [0117] Example 12: The system of any of the examples described herein, in particular example 11, wherein the first anchor comprises a bulky-knot. [0118] Example 13: The system of any one of any of the examples described herein, in particular examples 1 to 12, wherein the tether is configured to extend through the ventricular heart wall to couple to the second anchor. [0119] Example 14: A tether system comprising a first anchor configured to be coupled to a septal wall of a heart ventricle, a second anchor configured to be positioned over an externally oriented surface of a portion of a ventricular heart wall of the heart ventricle, and a tether configured to couple the first anchor and the second anchor to one another. [0120] Example 15: The system of any of the examples described herein, in particular example 14, wherein the first anchor is configured to be within the septal wall. [0121] Example 16: The system of any of the examples described herein, in particular example 14, wherein the first anchor is configured to be over a surface of the septal wall having an orientation opposite relative to that oriented toward the heart ventricle. [0122] Example 17: The system of any of the examples described herein, in particular examples 14 to 16, wherein the first anchor is configured to be coupled to a basal portion of the septal wall. [0123] Example 18: The system of any of the examples described herein, in particular examples 14 to 16, wherein the first anchor is configured to be coupled to a mid-portion of the septal wall. [0124] Example 19: The system of any of the examples described herein, in particular examples 14 to 18, wherein the second anchor is configured to be coupled to a mid-portion of the ventricular heart wall. [0125] Example 20: The system of any of the examples described herein, in particular examples 14 to 19, wherein the tether is configured to extend directly between the first anchor and the second anchor. [0126] Example 21: The system of any of the examples described herein, in particular examples 14 to 19, wherein the tether comprises a portion configured to form a loop around respective portions of papillary muscles and chordae tendineae in the heart ventricle. [0127] Example 22: The system of any of the examples described herein, in particular example 21, wherein the heart ventricle is a left ventricle, and wherein the portion of the tether comprises is configured to form a loop around both a posteromedial papillary muscle or chordae tendineae coupled to the posteromedial papillary muscle, and an anterolateral papillary muscle or chordae tendineae coupled to the posteromedial papillary muscle. [0128] Example 23: The system of any of the examples described herein, in particular examples 14 to 22, further comprising an elongate tube configured to receive at least a portion of the tether extending within the heart ventricle. [0129] Example 24: The system of any of the examples described herein, in particular examples 14 to 23, wherein the tether comprises a suture. [0130] Example 25: The system of any of the examples described herein, in particular example 24, wherein the first anchor comprises at least a portion formed using the suture. [0131] Example 26: The system of any of the examples described herein, in particular example 25, wherein the first anchor comprises a bulky-knot. [0132] Example 27: The system of any of the examples described herein, in particular examples 14 to 27, wherein the tether is configured to extend through the ventricular heart wall to couple to the second anchor. [0133] Example 28: A tether system comprising a first anchor configured to be coupled to a papillary muscle of a heart ventricle, a second anchor configured to be positioned over an externally oriented surface of an apical portion of a ventricular heart wall of the heart ventricle, and a tether configured to couple the first anchor and the second anchor to one another. [0134] Example 29: The system of any of the examples described herein, in particular example 28, wherein the heart ventricle is a left ventricle and the first anchor is configured to be coupled to an anterolateral papillary muscle. [0135] Example 30: The system of any of the examples described herein, in particular example 28 or 29, wherein the second anchor is configured to be coupled to an externally oriented surface of an apex of the ventricular heart wall. [0136] Example 31: The system of any of the examples described herein, in particular examples 28 to 30, further comprising an elongate tube configured to receive at least a portion of the tether extending within the heart ventricle. [0137] Example 32: The system of any of the examples described herein, in particular examples 28 to 31, wherein the tether comprises a suture. [0138] Example 33: The system of any of the examples described herein, in particular example 32, wherein the first anchor comprises at least a portion formed using the suture. [0139] Example 34: The system of any of the examples described herein, in particular example 33, wherein the first anchor comprises a bulky-knot. [0140] Example 35: The system of any of the examples described herein, in particular examples 28 to 34, wherein the tether is configured to extend through the ventricular heart wall to couple to the second anchor. [0141] Example 36: A system comprising a first anchor configured to be positioned over an externally oriented surface of a first portion of a ventricular heart wall of a heart ventricle, a second anchor configured to be positioned over an externally oriented surface of a second portion of the ventricular heart wall of the heart ventricle, and a tether configured to couple the first anchor and the second anchor to one another, the tether comprising a portion configured to form at least a partial loop around a papillary muscle of the heart ventricle or chordae tendineae coupled to the papillary muscle. [0142] Example 37: The system of any of the examples described herein, in particular example 36, wherein the first portion of the ventricular heart wall is above the second portion of the ventricular heart wall. [0143] Example 38: The system of any of the examples described herein, in particular example 36 or 37, wherein at least one of the first portion and the second portion are in a mid-portion of the ventricular heart wall. [0144] Example 39: The system of any of the examples described herein, in particular example 36 or 37, wherein at least one of the first portion and the second portion are in an apical portion of the ventricular heart wall. [0145] Example 40: The system of any one of any of the examples described herein, in particular examples 36 to 39, wherein the heart ventricle is a left ventricle and the portion of tether is configured to form at least a partial loop around a posteromedial papillary muscle or chordae tendineae coupled to the posteromedial papillary muscle. [0146] Example 41: The system of any of the examples described herein, in particular example 40, wherein the portion of the tether forms a partial loop around the posteromedial papillary muscle or chordae tendineae coupled to the posteromedial papillary muscle. [0147] Example 42: The system of any of the examples described herein, in particular examples 36 to 41, further comprising an elongate tube configured to receive at least a portion of the tether extending within the heart ventricle. [0148] Example 43: The system of any of the examples described herein, in particular examples 36 to 42, wherein the tether comprises a suture. [0149] Example 44: The system of any of the examples described herein, in particular example 43, wherein the first anchor comprises at least a portion formed using the suture. [0150] Example 45: The system of any of the examples described herein, in particular example 44, wherein the first anchor comprises a bulky-knot. [0151] Example 46: The system of any of the examples described herein, in particular examples 36 to 45, wherein the tether is configured to extend through the ventricular heart wall to couple to the second anchor. [0152] Example 47: A method comprising coupling a first anchor to an annulus of a heart valve that is between a heart atrium and a heart ventricle, extending a portion of a tether within the heart ventricle from the first anchor to a portion of a ventricular heart wall of the heart ventricle between an apex and a basal portion of the ventricular wall, positioning a second anchor over an externally oriented surface of the portion of the ventricular heart wall, and coupling the tether to the second anchor while the second anchor is over the externally oriented surface of the ventricular heart wall. [0153] Example 48: The method of any of the examples described herein, in particular example 47, wherein extending the portion of the tether from the first anchor to the portion of the heart ventricular heart comprises extending the tether from the first anchor to a mid-portion of the ventricular heart wall. [0154] Example 49: The method of any of the examples described herein, in particular example 47, wherein extending the portion of the tether from the first anchor to the portion of the ventricular heart wall comprises extending the tether from the first anchor to a portion of the ventricular heart wall between an apical portion and a mid-portion of the ventricular heart wall. [0155] Example 50: The method of any of the examples described herein, in particular example 47, wherein extending the portion of the tether from the first anchor to the portion of the ventricular heart wall comprises extending the tether from the first anchor to a portion of the ventricular heart wall adjacent to a papillary muscle. [0156] Example 51: The method of any of the examples described herein, in particular example 50, wherein extending the portion of the tether from the first anchor to the portion of the ventricular heart wall comprises extending the tether from the first anchor to a portion of the ventricular heart wall adjacent to anterolateral papillary muscle. [0157] Example 52: The method of any of the examples described herein, in particular examples 47 to 51, wherein coupling the first anchor to the annulus of the heart valve comprises coupling the first anchor to a mitral valve annulus. [0158] Example 53: The method of any of the examples described herein, in particular example 52, wherein coupling the first anchor to the annulus of the heart valve comprises coupling the first anchor to a portion of the mitral valve annulus adjacent to a posterior leaflet. [0159] Example 54: The method of any of the examples described herein, in particular example 52, wherein coupling the first anchor to the annulus of the heart valve comprises coupling the first anchor to a portion of the mitral valve annulus adjacent to an anterior leaflet. [0160] Example 55: The method of any of the examples described herein, in particular examples 47 to 54, further comprising advancing an elongate tube along at least a portion of the tether within the heart ventricle.

[0161] The above method(s) can be performed on a living animal or on a simulation, such as on a cadaver, cadaver heart, anthropomorphic ghost, simulator (e.g., with body parts, heart, tissue, etc. being simulated). [0162] Example 56: A method comprising coupling a first anchor to a septal wall of a heart ventricle, extending a portion of a tether within the heart ventricle from the first anchor to a portion of a ventricular heart wall of the heart ventricle, positioning a second anchor over an externally oriented surface of the portion of the ventricular heart wall, and coupling the tether to the second anchor while the second anchor is over the externally oriented surface of the ventricular heart wall. [0163] Example 57: The method of any of the examples described herein, in particular example 56, wherein coupling the first anchor to the septal wall comprises embedding the first anchor within the septal wall. [0164] Example 58: The method of any of the examples described herein, in particular example 56, wherein coupling the first anchor to the septal wall comprises positioning the first anchor over a surface of the septal wall having an opposing orientation relative to that oriented toward the heart ventricle. [0165] Example 59: The method of any of the examples described herein, in particular examples 56 to 58, wherein coupling the first anchor to the septal wall comprises coupling the first anchor to a basal portion of the septal wall. [0166] Example 60: The method of any of the examples described herein, in particular examples 56 to 58, wherein coupling the first anchor to the septal wall comprises coupling the first anchor to a mid-portion of the septal wall. [0167] Example 61: The method of any of the examples described herein, in particular examples 56 to 60, wherein extending the portion of the tether from the first anchor to the portion of the ventricular heart wall of the heart ventricle comprises extending the portion of the tether from the first anchor to a mid-portion of the ventricular heart wall. [0168] Example 62: The method of any of the examples described herein, in particular examples 56 to 61, wherein extending the portion of the tether from the first anchor to the portion of the ventricular heart wall of the heart ventricle comprises extending the portion of the tether directly from the first anchor to the second anchor. [0169] Example 63: The method of any of the examples described herein, in particular examples 56 to 61, wherein extending the portion of the tether from the first anchor to the portion of the ventricular heart wall of the heart ventricle comprises looping the tether around both a posteromedial papillary muscle or chordae tendineae coupled to the posteromedial papillary muscle, and an anterolateral papillary muscle or chordae tendineae coupled to the posteromedial papillary muscle. [0170] Example 64: The method of any of the examples described herein, in particular examples 56 to 63, further comprising advancing an elongate tube along at least a portion of the tether within the heart ventricle.

[0171] The above method(s) can be performed on a living animal or on a simulation, such as on a cadaver, cadaver heart, anthropomorphic ghost, simulator (e.g., with body parts, heart, tissue, etc. being simulated). [0172] Example 65: A method comprising coupling a first anchor to a papillary muscle of a heart ventricle, extending a portion of a tether within the heart ventricle from the first anchor to an apical portion of a ventricular heart wall of the heart ventricle, positioning a second anchor over an externally oriented surface of the apical portion of the ventricular heart wall, and coupling the tether to the second anchor while the second anchor is over the externally oriented surface of the apical portion of the ventricular heart wall. [0173] Example 66: The method of any of the examples described herein, in particular example 65, wherein coupling the first anchor to the papillary muscle comprises coupling the first anchor to an anterolateral papillary muscle. [0174] Example 67: The method of any of the examples described herein, in particular example 65 or 66, wherein extending the portion of the tether within the heart ventricle from the first anchor to the apical portion of the ventricular heart wall comprises extending the portion of the tether within the heart ventricle from the first anchor to an apex of the ventricular heart wall. [0175] Example 68: The method of any of the examples described herein, in particular examples 65 to 67, further comprising advancing an elongate tube along at least a portion of the tether within the heart ventricle.

[0176] The above method(s) can be performed on a living animal or on a simulation, such as on a cadaver, cadaver heart, anthropomorphic ghost, simulator (e.g., with body parts, heart, tissue, etc. being simulated). [0177] Example 69: A method comprising coupling a first anchor to a first portion of a ventricular heart wall of a heart ventricle, and extending a portion of a tether within the heart ventricle from the first anchor to a second portion of the ventricular heart wall of the heart ventricle and forming at least a partial loop, using the tether, around a papillary muscle of the heart ventricle or chordae tendineae coupled to the papillary muscle. The method can include positioning a second anchor over an externally oriented surface of the second portion of the ventricular heart wall, and coupling the tether to the second anchor while the second anchor is over the externally oriented surface of the ventricular heart wall. [0178] Example 70: The method of any of the examples described herein, in particular example 69, wherein extending the portion of the tether within the heart ventricle from the first anchor to the second portion of the ventricular heart wall comprises extending the portion of the tether within the heart ventricle from the first anchor to a second portion of the ventricular heart wall that is lower on the ventricular heart wall than the first portion. [0179] Example 71: The method of any of the examples described herein, in particular example 69 or 70, wherein coupling the first anchor to the first portion of the ventricular heart wall comprises coupling the first anchor to a first mid-portion of the ventricular heart wall. [0180] Example 72: The method of any of the examples described herein, in particular example 71, wherein extending the portion of the tether within the heart ventricle from the first anchor to the second portion of the ventricular heart wall comprises extending the portion of the tether within the heart ventricle from the first anchor to a second mid-portion of the ventricular heart wall. [0181] Example 73: The method of any of the examples described herein, in particular example 71, wherein extending the portion of the tether within the heart ventricle from the first anchor to the second portion of the ventricular heart wall comprises extending the portion of the tether within the heart ventricle from the first anchor to a second apical portion of the ventricular heart wall. [0182] Example 74: The method of any of the examples described herein, in particular example 69, wherein: [0183] coupling the first anchor to the first portion of the ventricular heart wall comprises coupling the first anchor to a first apical portion of the ventricular heart wall; and [0184] extending the portion of the tether within the heart ventricle from the first anchor to the second portion of the ventricular heart wall comprises extending the portion of the tether within the heart ventricle from the first anchor to a second apical portion of the ventricular heart wall. [0185] Example 75: The method of any of the examples described herein, in particular examples 69 to 75, wherein forming at least a partial loop, using the tether, around the papillary muscle comprises forming at least a partial loop, using the tether, around a posteromedial papillary muscle or chordae tendineae coupled to the posteromedial papillary muscle. [0186] Example 76: The method of any of the examples described herein, in particular examples 69 to 75, wherein forming at least a partial loop, using the tether, around the posteromedial papillary muscle comprises forming a partial loop around the posteromedial papillary muscle or chordae tendineae coupled to the posteromedial papillary muscle. [0187] Example 77: The method of any of the examples described herein, in particular examples 69 to 76, further comprising advancing an elongate tube along at least a portion of the tether within the heart ventricle.

[0188] The above method(s) can be performed on a living animal or on a simulation, such as on a cadaver, cadaver heart, anthropomorphic ghost, simulator (e.g., with body parts, heart, tissue, etc. being simulated).

[0189] Depending on the example, certain acts, events, or functions of any of the processes or algorithms described herein can be performed in a different sequence, may be added, merged, or left out altogether. Thus, in certain examples, not all described acts or events are necessary for the practice of the processes.

[0190] Conditional language used herein, such as, among others, “can,” “could,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is intended in its ordinary sense and is generally intended to convey that certain examples include, while other examples do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more examples or that one or more examples necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular example. The terms “comprising,” “including,” “having,” and the like are synonymous, are used in their ordinary sense, and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Conjunctive language such as the phrase “at least one of X, Y and Z,” unless specifically stated otherwise, is understood with the context as used in general to convey that an item, term, element, etc. may be either X, Y or Z. Thus, such conjunctive language is not generally intended to imply that certain examples require at least one of X, at least one of Y and at least one of Z to each be present.

[0191] It should be appreciated that in the above description of examples, various features are sometimes grouped together in a single example, Figure, or description thereof for the purpose of streamlining the disclosure and aiding in the understanding of one or more of the various inventive aspects. This method of disclosure, however, is not to be interpreted as reflecting an intention that any claim require more features than are expressly recited in that claim. Moreover, any components, features, or steps illustrated and/or described in a particular example herein can be applied to or used with any other example(s). Further, no component, feature, step, or group of components, features, or steps are necessary or indispensable for each example. Thus, it is intended that the scope of the inventions herein disclosed and claimed below should not be limited by the particular examples described above, but should be determined only by a fair reading of the claims that follow.

[0192] It should be understood that certain ordinal terms (e.g., “first” or “second”) may be provided for ease of reference and do not necessarily imply physical characteristics or ordering. Therefore, as used herein, an ordinal term (e.g., “first,” “second,” “third,” etc.) used to modify an element, such as a structure, a component, an operation, etc., does not necessarily indicate priority or order of the element with respect to any other element, but rather may generally distinguish the element from another element having a similar or identical name (but for use of the ordinal term). In addition, as used herein, indefinite articles (“a” and “an”) may indicate “one or more” rather than “one.” Further, an operation performed “based on” a condition or event may also be performed based on one or more other conditions or events not explicitly recited.

[0193] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example examples belong. It be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0194] The spatially relative terms “outer,” “inner,” “upper,” “lower,” “below,” “above,” “vertical,” “horizontal,” and similar terms, may be used herein for ease of description to describe the relations between one element or component and another element or component as illustrated in the drawings. It be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation, in addition to the orientation depicted in the drawings. For example, in the case where a device shown in the drawing is turned over, the device positioned “below” or “beneath” another device may be placed “above” another device. Accordingly, the illustrative term “below” may include both the lower and upper positions. The device may also be oriented in the other direction, and thus the spatially relative terms may be interpreted differently depending on the orientations.

[0195] Unless otherwise expressly stated, comparative and/or quantitative terms, such as “less,” “more,” “greater,” and the like, are intended to encompass the concepts of equality. For example, “less” can mean not only “less” in the strictest mathematical sense, but also, “less than or equal to.”

Claims

1. A method comprising: coupling a first anchor to an annulus of a heart valve that is between a heart atrium and a heart ventricle; extending a portion of a tether within the heart ventricle from the first anchor to a portion of a ventricular heart wall of the heart ventricle between an apex and a basal portion of the ventricular heart wall; positioning a second anchor over an externally oriented surface of the portion of the ventricular heart wall; and coupling the tether to the second anchor while the second anchor is over the externally oriented surface of the ventricular heart wall.

2. The method of claim 1, wherein extending the portion of the tether from the first anchor to the portion of the ventricular heart wall comprises extending the tether from the first anchor to a mid-portion of the ventricular heart wall.

3. The method of claim 1, wherein extending the portion of the tether from the first anchor to the portion of the ventricular heart wall comprises extending the tether from the first anchor to a portion of the ventricular heart wall between an apical portion and a mid-portion of the ventricular heart wall.

4. The method of claim 1, wherein extending the portion of the tether from the first anchor to the portion of the ventricular heart wall comprises extending the tether from the first anchor to a portion of the ventricular heart wall adjacent to a papillary muscle.

5. The method of claim 4, wherein extending the portion of the tether from the first anchor to the portion of the ventricular heart wall comprises extending the tether from the first anchor to a portion of the ventricular heart wall adjacent to anterolateral papillary muscle.
6. The method of claim 1, wherein coupling the first anchor to the annulus of the heart valve comprises coupling the first anchor to a mitral valve annulus.
7. The method of claim 6, wherein coupling the first anchor to the annulus of the heart valve comprises coupling the first anchor to a portion of the mitral valve annulus adjacent to a posterior leaflet.
8. The method of claim 6, wherein coupling the first anchor to the annulus of the heart valve comprises coupling the first anchor to a portion of the mitral valve annulus adjacent to an anterior leaflet.
9. A method comprising: anchoring a first portion of a tether to a portion of an annulus of a heart valve that is between a heart atrium and a heart ventricle; extending a medial portion of the tether from the portion of the annulus through the heart ventricle to a portion of a ventricular heart wall of the heart ventricle between an apex and a basal portion of the ventricular heart wall; and anchoring a second portion of the tether to the portion of the ventricular heart wall to pull the portion of the ventricular heart wall toward the portion of the annulus and provide reversal in remodeling of the ventricular heart wall.
10. The method of claim 9, wherein extending the medial portion of the tether comprises extending the medial portion of the tether from the portion of the annulus through the heart ventricle to a mid-portion of the ventricular heart wall.
11. The method of claim 9, wherein extending the medial portion of the tether comprises extending the medial portion of the tether from the portion of the annulus through the heart ventricle to a portion of the ventricular heart wall between an apical portion and a mid-portion of the ventricular heart wall.
12. The method of claim 9, wherein extending the medial portion of the tether comprises extending the medial portion of the tether from the portion of the annulus through the heart ventricle to a portion of the ventricular heart wall adjacent to a papillary muscle to avoid traversing areas populated by papillary muscles.
13. The method of claim 9 further comprising advancing an elongate tube along at least a portion of the medial portion of the tether to prevent contact between the at least a portion of the medial portion of the tether and heart tissue.
14. A method comprising: providing an anchor over an atrium-facing surface of an annulus of a heart valve; tethering the anchor through a heart ventricle to a portion of a ventricular heart wall of the heart ventricle between an apex and a basal portion of the ventricular heart wall; and anchoring the tether to the portion of the ventricular heart wall.
15. The method of claim 14, wherein providing the anchor over the atrium-facing surface of the annulus comprises forming the anchor using a portion of a suture.
16. The method of claim 15, wherein forming the anchor using the portion of the suture comprises forming a bulky-knot using the portion of the suture.
17. The method of claim 15, wherein tethering the anchor through the heart ventricle to the portion of the ventricular heart wall comprises extending another portion of the suture through the annulus and the heart ventricle to the portion of the ventricular heart wall.
18. The method of claim 14, wherein tethering the anchor through the heart ventricle to the portion of the ventricular heart wall comprises tethering the anchor through the heart ventricle to a mid-portion of the ventricular heart wall.
19. The method of claim 14, wherein tethering the anchor through the heart ventricle to the portion of the ventricular heart wall comprises tethering the anchor through the heart ventricle to a portion of the ventricular heart wall between an apical portion and a mid-portion of the ventricular heart wall.

20. The method of claim 14, wherein tethering the anchor through the heart ventricle to the portion of the ventricular heart wall comprises tethering the anchor through the heart ventricle to a portion of the ventricular heart wall adjacent to a papillary muscle.